DISTRIBUTION RELIGION

THE IMAGE PROCESSOR MAY BE COPIED BY INDIVIDUALS AND NOT-FOR-PROFIT INSTITUTIONS WITHOUT CHARGE. FOR-PROFIT INSTITUTIONS WILL HAVE TO NEGOTIATE FOR PERMISSION TO COPY. I THINK CULTURE HAS TO LEARN TO USE HIGH-TEK MACHINES FOR PERSONAL AESTHETIC, RELIGIOUS, INTUITIVE, COMPREHENSIVE, EXPLORATORY GROWTH. THE DEVELOPMENT OF MACHINES LIKE THE IMAGE PROCESSOR IS PART OF THIS EVOLUTION. I AM PAID BY THE STATE, AT LEAST IN PART, TO DO AND DISEMINATE THIS INFORMATION; SO I DO.

As I am sure you (who are you) understand a work like developing and expanding the Image Processor requires much money and time. The'U' does not have much money for evolutionary work and getting of grants are almost as much work as holding down a job. Therefore, I have the feeling that if considerable monies were to be made with a copy of the Image Processor, I would like some of it.

Put in your own method of returning energy to me here: _____

_____ _____

_____ _____

_____ _____

_____ _____

_____ _____

_____ _____

Of course enforcing such a request is too difficult to be bothered with. But let it be known that I consider it to be morally binding.

Much Love,

I DECIDED THAT I WOULD LIKE 1 good TAPE FROM EACH COPY of THE I. P.

Daniel J. Sandin
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Lab phone: 312-996-2312
Messages: 312-996-3337 (Department of Art)

NOTES ON THE AESTHETICS OF 'copying-an-Image Processor':

Being a 'copier of many things, in this case the first copier of an Image Processor, I trust the following notes to find meaning to future copiers of Image Processors:

First, it's okay to copy! Believe in the process of copying as much as you can; with all your heart is a good place to start - get into it as straight and honestly as possible. Copying is as good (I think better from this vector-view) as any other way of getting 'there.'

The more you 'buy' the 'copying' of Sandin's encoded intelligence in the I-P, the more you will learn about the man-and-machines. Don't try to make improvements; you'll make it only worse if you modify what already is best, even if it doesn't appear to be the 'best' to your mind's eye. It bothers me very much to see 'folk' laying onto Dan, suggestions of improvement (supposedly) without a thorough giving-in-to understanding of the I-P design. Please realize, that if you 'had-it' to do it you would not be building (copying) an I-P to begin with; you would have done it yourself along time ago...so get to work copying--as-usual.

Dan's evolutionary design of the I-P comes from a very high and thorough CONSCIOUS systems--design-intelligence-level. If you deviate in the process of 'copying' and then Dan makes an improvement on his I-P, you will most likely find it quite frustrating in updating your instrument due to your I-P being incompatible in detail to the original. If you get yourself in a jam, then you have to go to Dan and "$PEND' his time getting you out of it.

So...after all this: the Art of 'copying' is a good form to try on for a year or so while you get into building your Image Processor...enjoy.

PEACE/ASCESIS (love):

Phil Morton

OCT 78

~~FEBRUARY 20, 1976~~

This edition of the documentation was paid for by a grant from the Illinois Arts Council. Thank you, Ill. Arts!

HI:

A bunch of miscelaneous notes -

If you didn't send postage, send it in stamps, money or check, or any thing else of comparable value (*surprise has intrinsic value*); postage costs me $2.00.

AND COPPYS OF (IN FORMATION and) COVECTIONS COST ME 1$

The master parts list contains the minimum order to complete the Image Processor. It is necessary to order more than the minimum of nearly everything. Parts may be damaged in assembly or may be defective. Although the Image Processor is very reliable, replacement parts are necessary for maintence. Furthermore, I attempt to design with a minimum of different parts, therefore new modules or modifications of modules are likely to use the same parts. With the exception of the hardware and the most expensive components, I reccommend ordering many extra.

If you need clarification on details; CALL (or send video tape). Don't write; I hate to write.

STILL TRUE IN 78

New corrections and additions are forth-coming in a few months. When ready to build, send self addressed stamped envelope ~~(5oz. should do)~~ ? Mention the last date of corrections you have.

CORRECTION OF ERRORS IN DOCUMENTATION

Master Parts List:

The following-

1000 FT	36F1800WM RG 59/U	CO-AXIAL CABLE	81.28	81.28	N

SHOULD BE CHANGED TO

1000 ft	RG 59/U COAXIAL CABLE		50.85	50.85	N
36F 110WM	(BELDEN # 8241-1000)				

PAGES FOLOWING DOCUMENTATION ERRORS SHOULD REPLACE CORISPONDING PAGES IN THE DOCUMENTATION

BRIEF SYSTEMS LEVEL DESCRIPTION:

The IP physically is an array of a minimum of approximately 24 modules (aluminum boxes), representing approximately 40 electrical modules.

The documentation that follows is simply a description of how to build the aluminum boxes; the system is considerably more powerful than the sum of the boxes.

On paper a description of how the IP works is more difficult than I am prepared to do. It is best communicated on video-tape; send me a video tape of you best stuff and I will send you a video tape on the IP, and/or send blank tape and $5/hr. (2 hours should do it.)

But in brief, the Image Processor accepts signals = $\pm$ .5 volts 75 ohm including video signals. These signals (images) are distributed into (usually) a number of processing modules and then (usually) mixed out into a standard color encoder (output module). Since most of the processing modules are voltage controllable and control voltages and images are interchangeable, fantastic combinatorial power is possible.

The 'classic' Image Processor contains 8 adder-multipliers, 3 function generators, 3 comparators, 3 amplitude classifiers, 4 oscillators, 3 differentiators, 9 references, 1 sync strip and camera input, 3 inputs, 1 sync generator, 1 color encoder and power supplies. These refer to electrical modules and not aluminum boxes. This constitutes a very powerful processing instrument and because of systems power level (inter-connect-ability), I recommend building approximately this much.

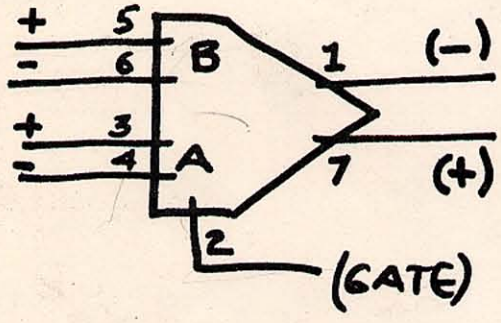
MC 1445 Gain Controlled Amplifier (multiplier):

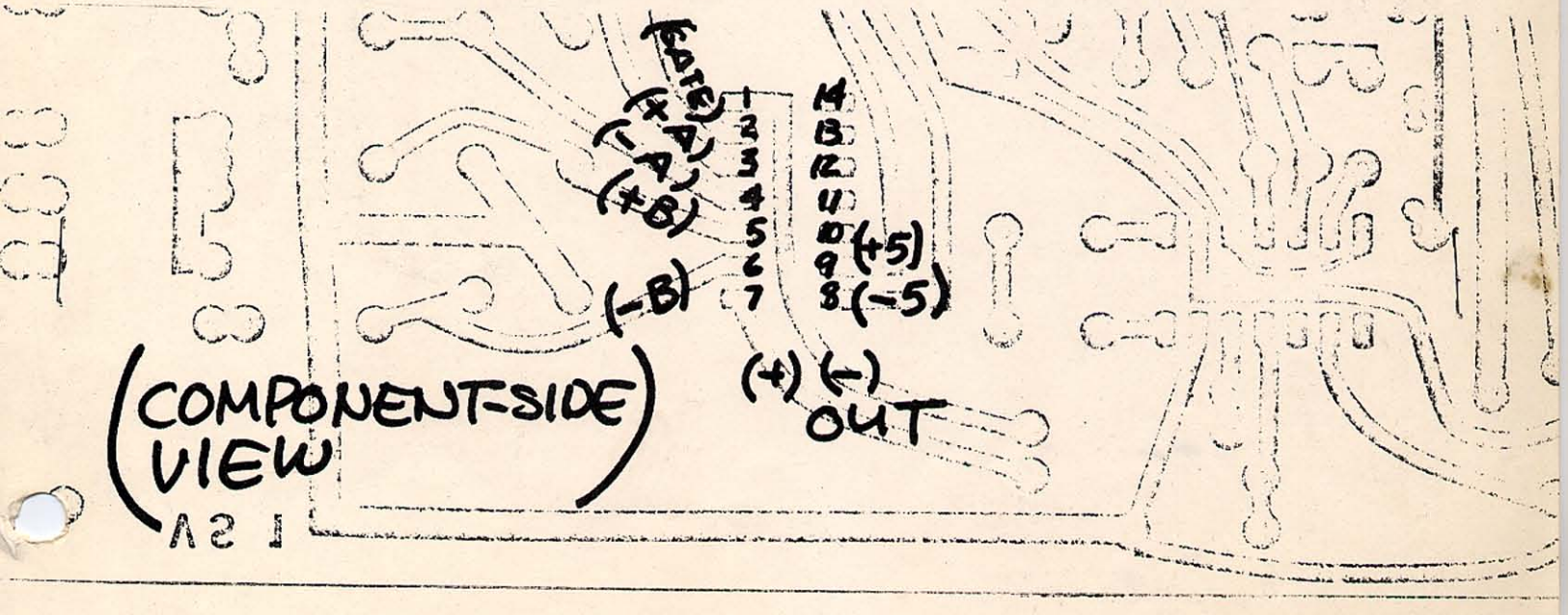
Detailed information on this integrated circuit is available from Motorola Linear Integrated Circuits Manual, available from Motorola or some Motorola distributors.

This I.C. is a four input gain controlled amplifier and is used throughout the IP. Pin (4) is the inverting input to channel A. Pin (3) is the non-inverting input to channel A. Pin (5) is the non-inverting input to channel B; pin (6) is the inverting input to channel B. Non-inverting output is available at pin (7); and the inverting output is available at pin (1).

Which input channel is connected to the output is controlled by the gate voltage at pin (2). If this voltage is high (greater than 1 volt) channel B is on; if the gate voltage is low (0 volts) channel A is connected to the output. The gate voltage produces continous control over the gains of the channels such that .5 volts causes both channel A and B to be connected to the output with 1/2 gain each. Full gain is approximately 10.

Power supply voltage (+5 volts) is connected to pin (9) and (-5 volts) is connected to pin (8). No other pins are used.



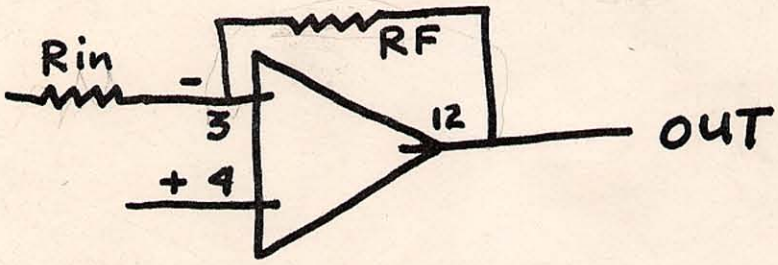


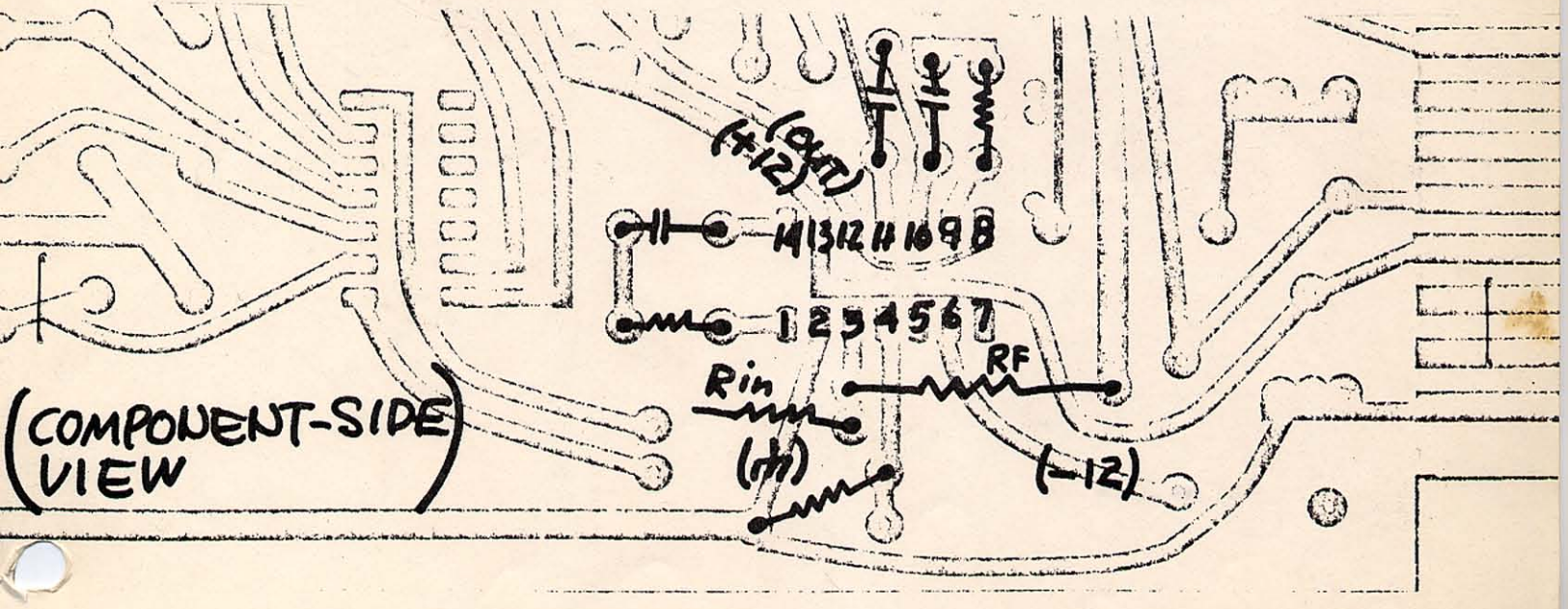
CA 3030 operational amplifier:

The CA 3030 op amp is used through out the IP. Detailed descriptions of the device are available in the RCA Linear Integrated Circuits Manuel. The book can be gotten from RCA or some distributors of RCA integrated circuits.

What follows is a brief description of the I.C.
The Op Amp has a very large gain (4000). Except in the comparator circuit, this gain is reduced by feedback of a percentage of the output signal pin (12) to the inverting input signal pin (3). A signal to be amplified is applied to pin (3) and will be inverted in the output, or it is applied to pin (4) and is not inverted. Pin (2) is grounded always. Pins (1,14,9,10,11) have to do with compensation for the amplifier which controls the tendency of the amplifier to oscillate (put out a signal of its own). The positive supply voltage (+12v.) is applied to pin (13); negative supply voltage (-12v.) is applied to pin (6). Pin (8) is sometimes connected to the output pin (12) to increase the power available from the I.C.

In simple inverting amplifier circuits, the voltage gain of the amplifier is the ratio of the feedback resistor between pin (12) and pin (3) to the input resistor connected from the input signal to pin (3).
RF/R in

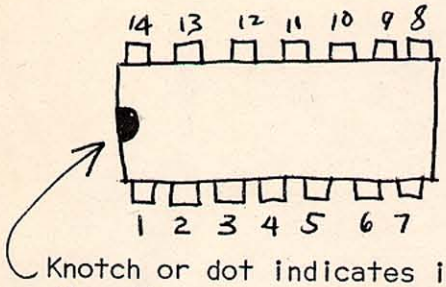




INTEGRATED CIRCUITS

Dual Inline Packages (DIP)

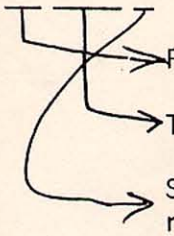
TOP VIEW:



Knotch or dot indicates installation direction.

IDENTIFICATION NO. (example):

MC 1445 L



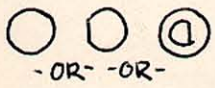
Prefix indicates manufacturer.

Type.

Suffix indicates case or temperature range or detail specifications.

TRANSISTORS

TOP VIEW:



Physical cases will vary with the manufacturer.

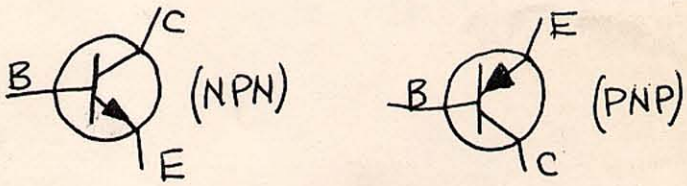
Check and match carefully the emitter, base, collector (EBC) leads with the NPN & PNP character of each transistor type.

Leads typically have to be bent for proper and convenient insertion into PC Board.

IDENTIFICATION NO.:
(only two transistors used in entire system)

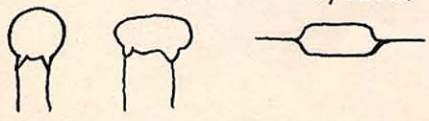
2N 4123 (NPN)
2N 4125 (PNP)

SYMBOL:



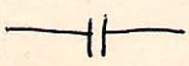
CAPACITORS

Ceramic Mica Polyestor

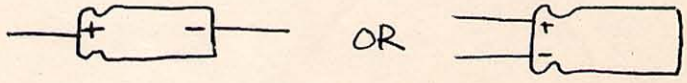


Installation direction makes no difference.

SYMBOL:

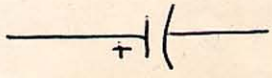


Electrolytic



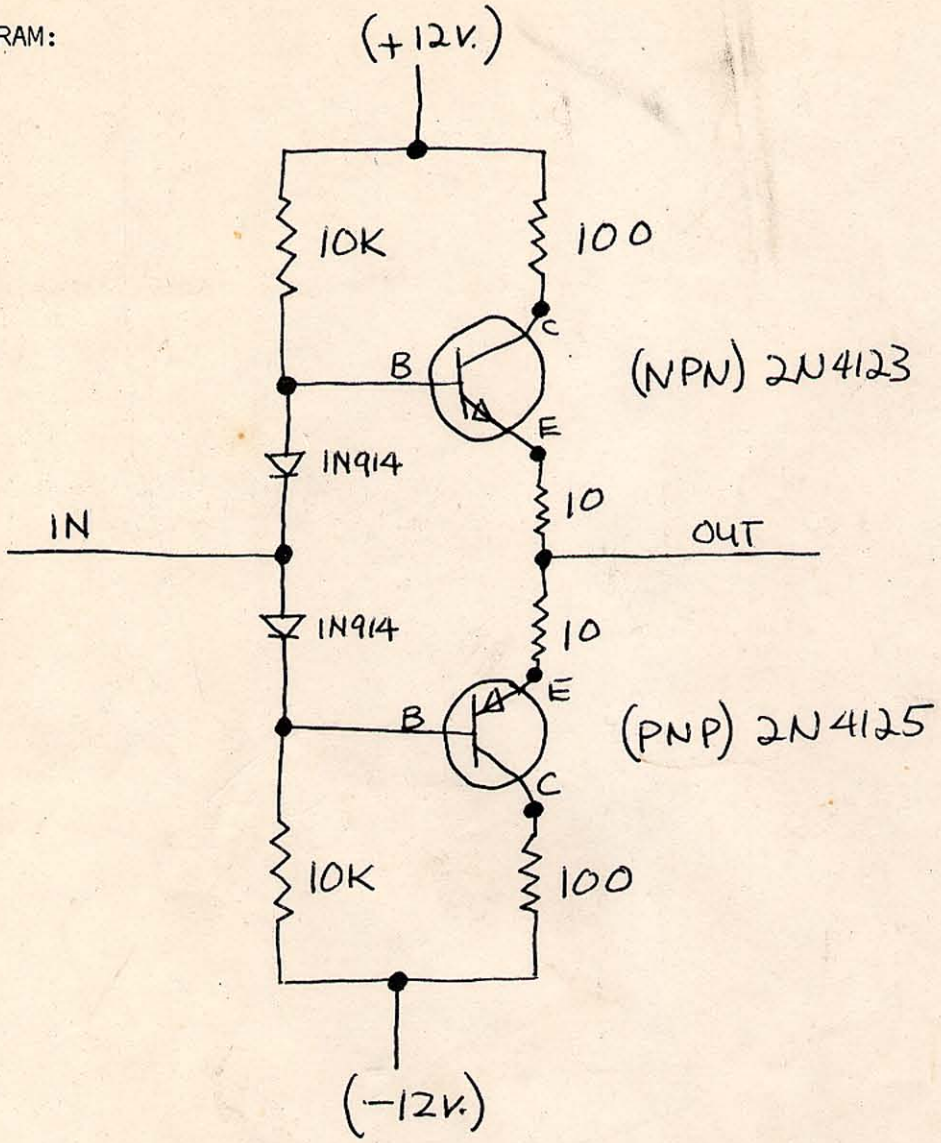
Installation must have correct (+) and (-) orientation.

SYMBOL:

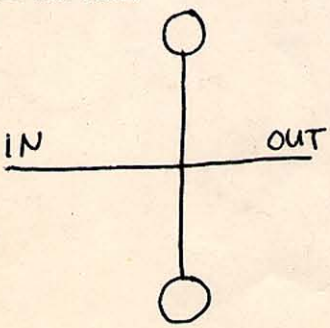


The STANDARD DRIVER is a complimentary current amplifier with voltage gain less than 1. It is used so many times in the I-P that it is abreviated:

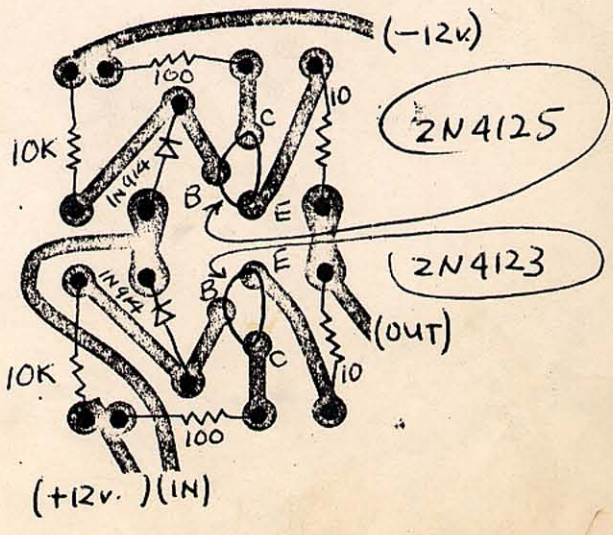
DIAGRAM:



ABREVIATION:

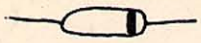


COMPONENT SIDE VIEW:



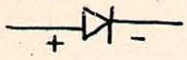
DIODES

COMPONENT:



Band or dot indicates the cathode (-).

SYMBOL:



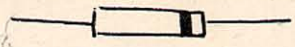
Direction of current flow is from (-) to (+).

IDENTIFICATION NO.:
(only two signal diodes used in entire system)

1N 914
1N 270

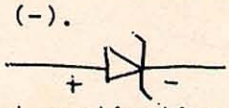
ZENER DIODES

COMPONENT:



Band or dot indicates the cathode (-).

SYMBOL:



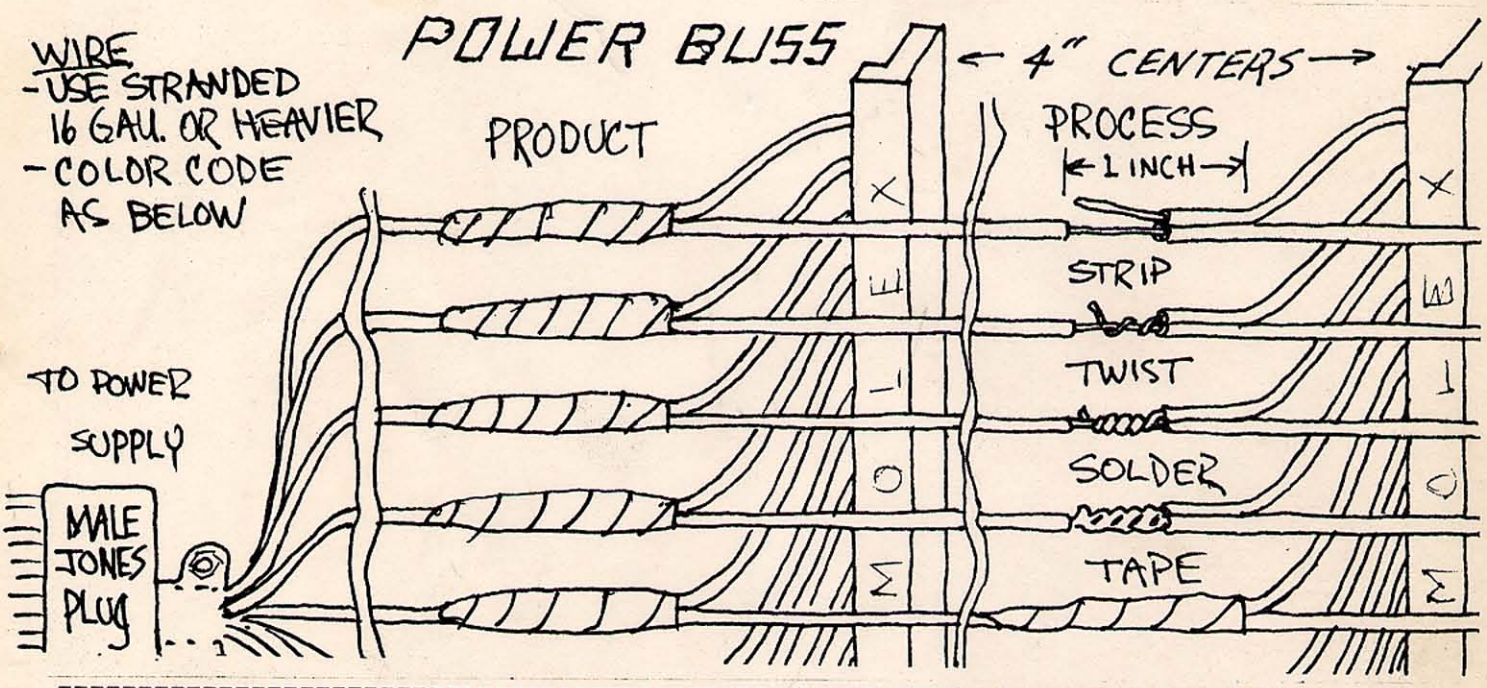
In normal application (+) supply voltage is supplied to cathode and (-) supply voltage is supplied to anode.

IDENTIFICATION NO. (example):

1N 5338 B

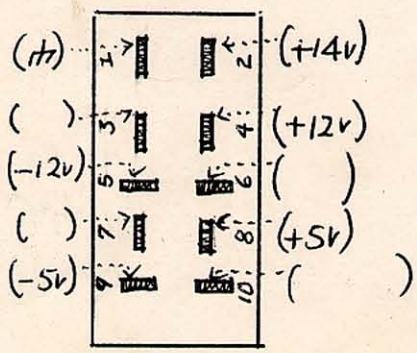
Indicates tolerance.

Type.

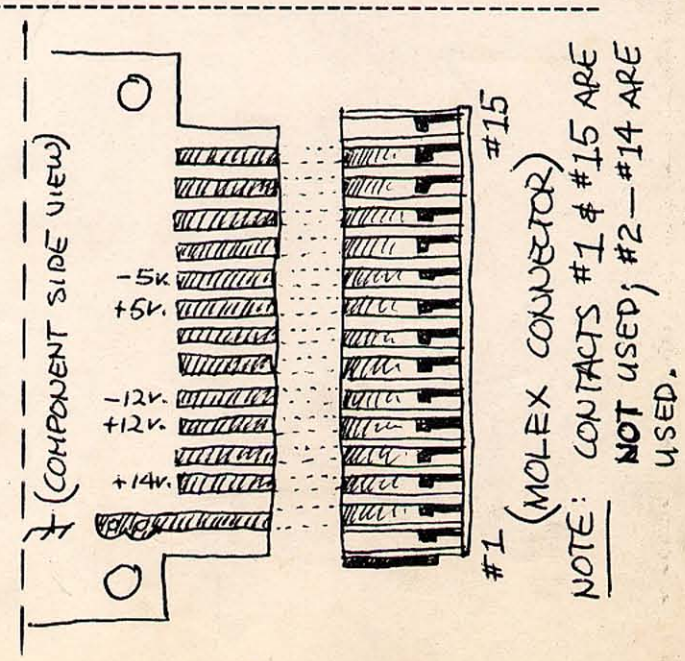


COLOR CODE FOR POWER BUSSING

BLACK (⏚)
ORANGE ( )
WHITE ( )
RED (+12v)
BLUE (-12v)
GREEN (+6v)
GRAY ( )
YELLOW (+5v)
VIOLET (-5v)
BROWN ( )

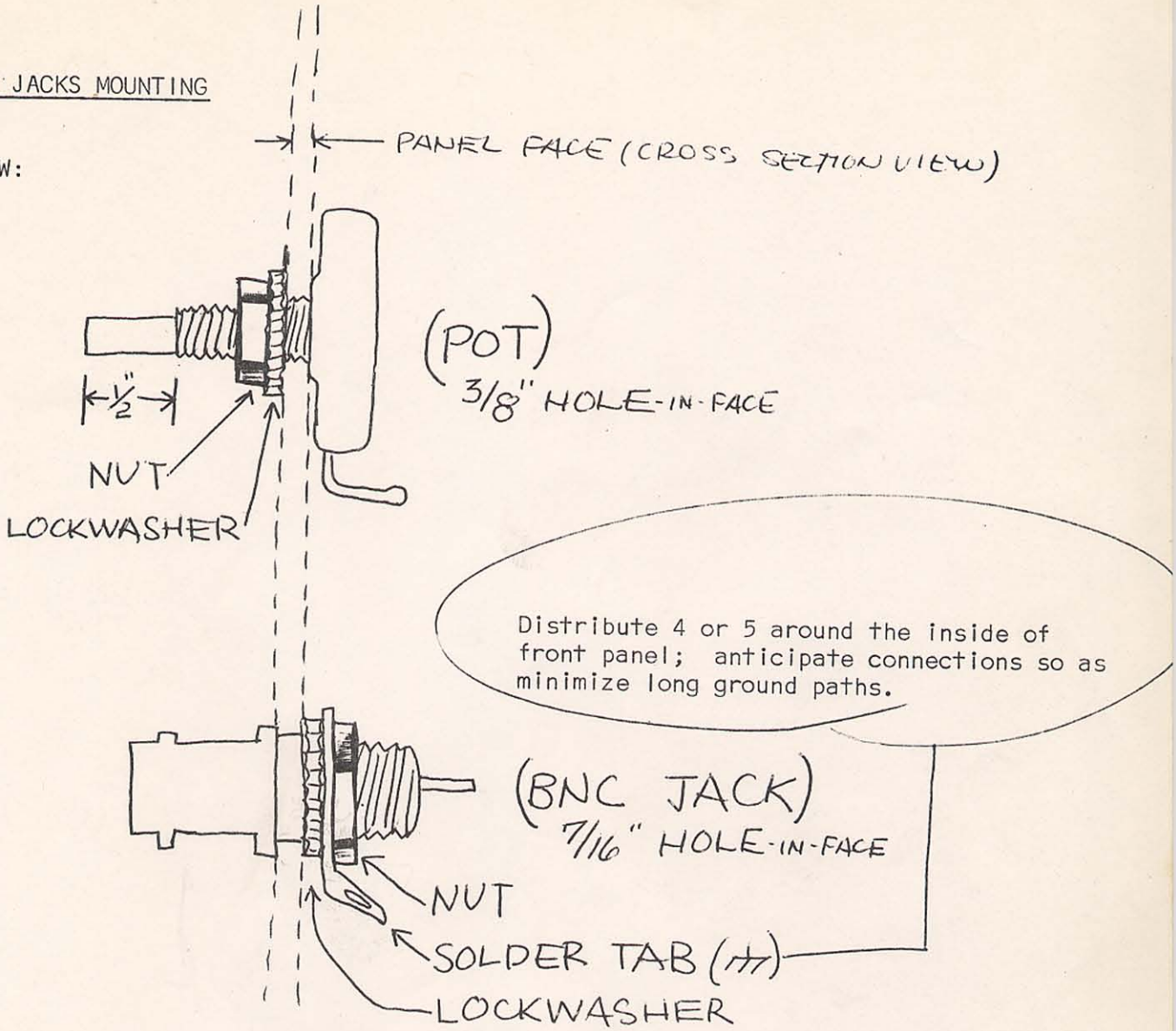


NOTE: All power supply lines into PC board are by-passed to ground (⏚) with a 100μF 25wvdc electrolytic capacitor (indicated in pictorials only).

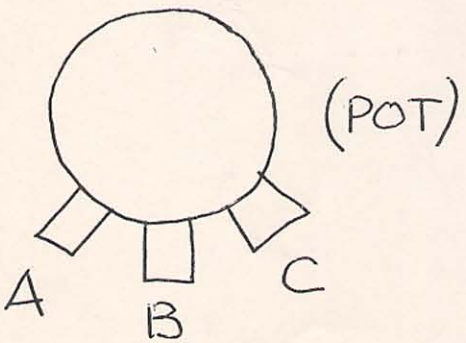


POTS and JACKS MOUNTING

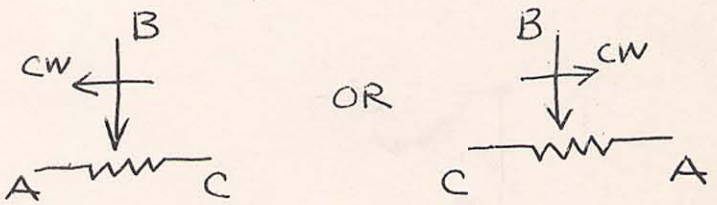
SIDE VIEW:



BACK VIEW:

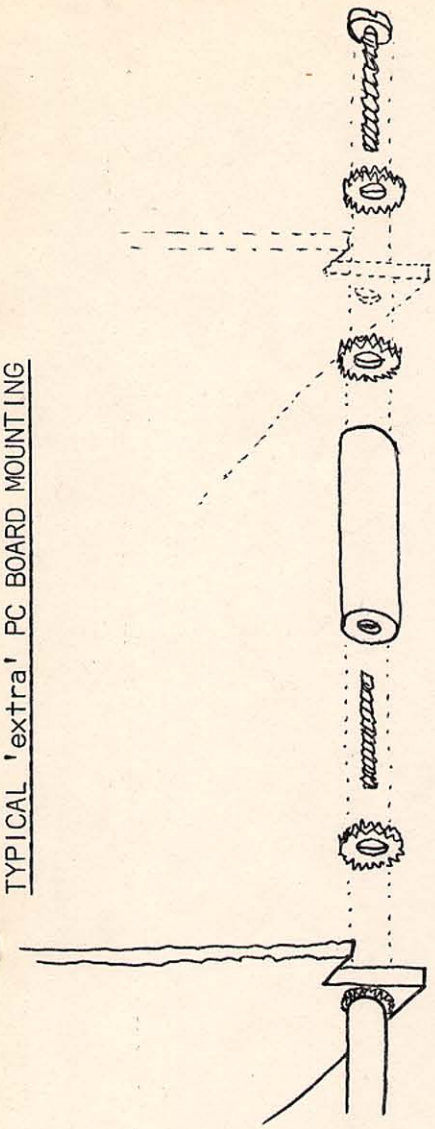


SYMBOL:



CW = clockwise

TYPICAL 'extra' PC BOARD MOUNTING



4-40 screw

star washer

extra PC board

star washer

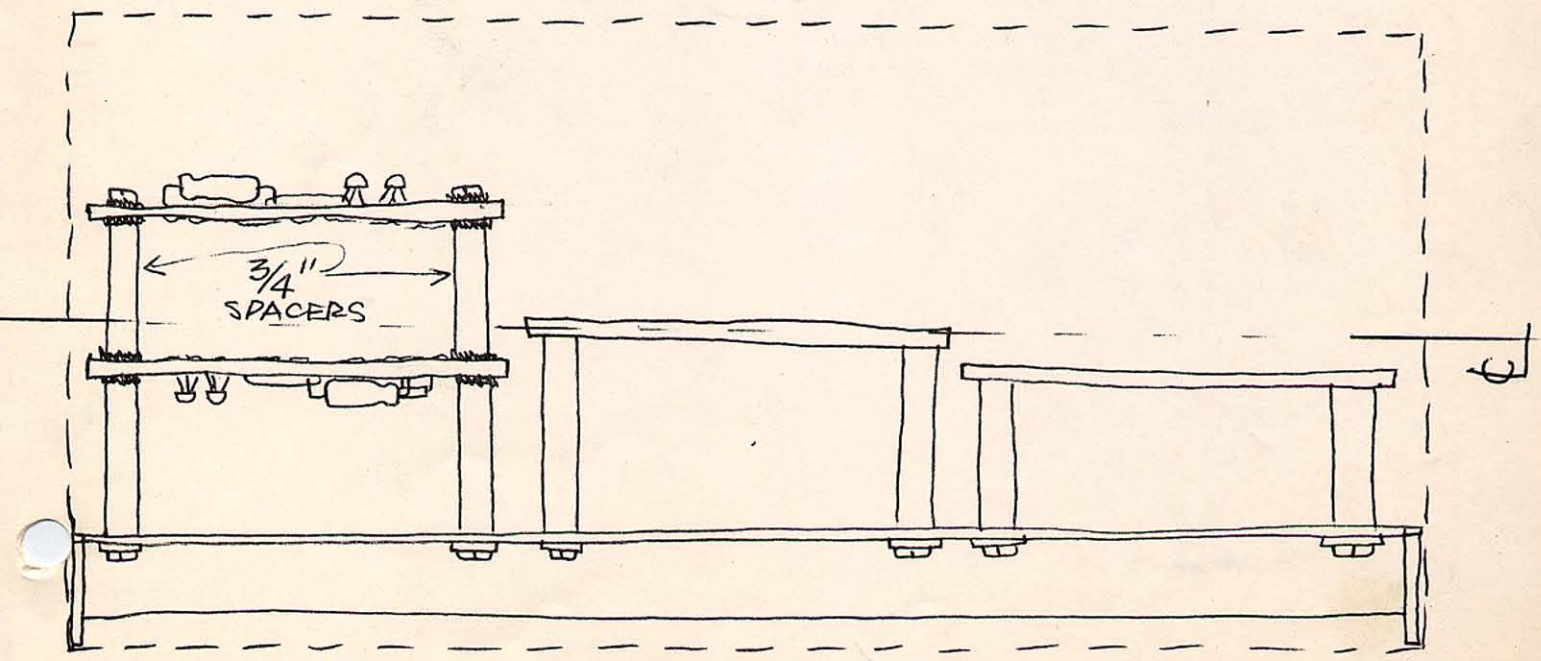
3/4" spacer

4-40 screw with head cut-off
making a 'threaded-shaft' screw-able
into both spacers

star washer

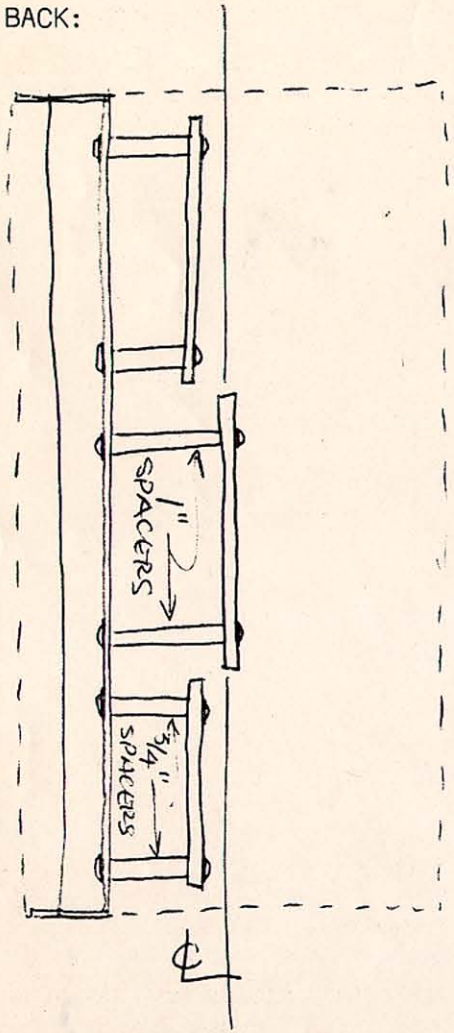
PC board

NOTE: Use additional washers for proper spacing when necessary; make sure that middle board is dead-on with center line of module.



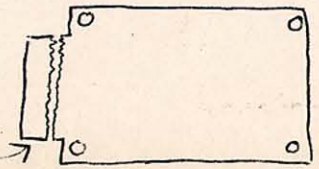
PRINTED CIRCUIT BOARD MOUNTING

BACK:

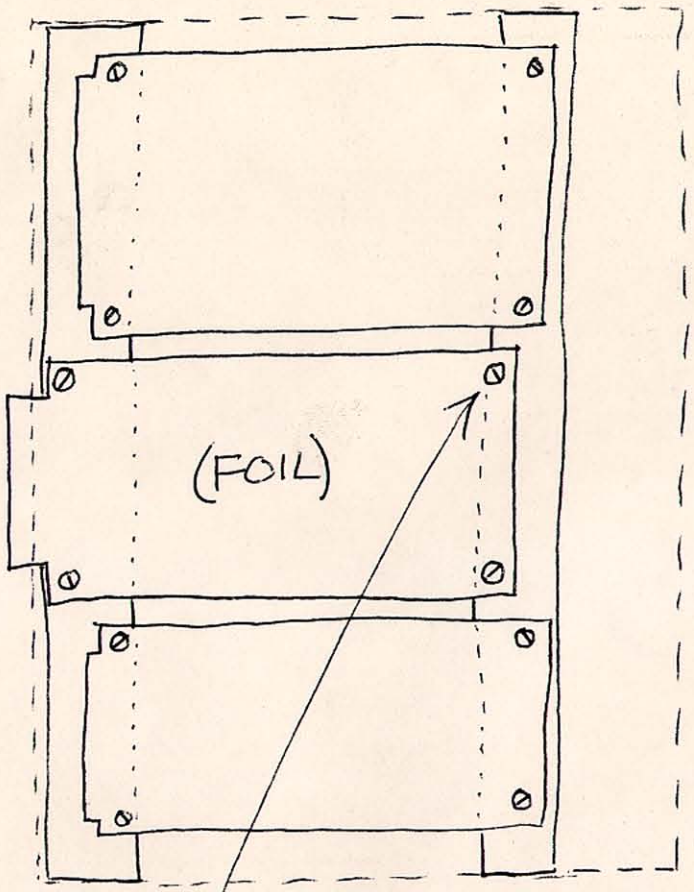


Adjust middle board so as to be dead-on with center line of module; use washers for additional spacing if neccessary.

On top and bottom boards it is necessary to hacksaw off 3/8" on power buss end for clearance of back panel.

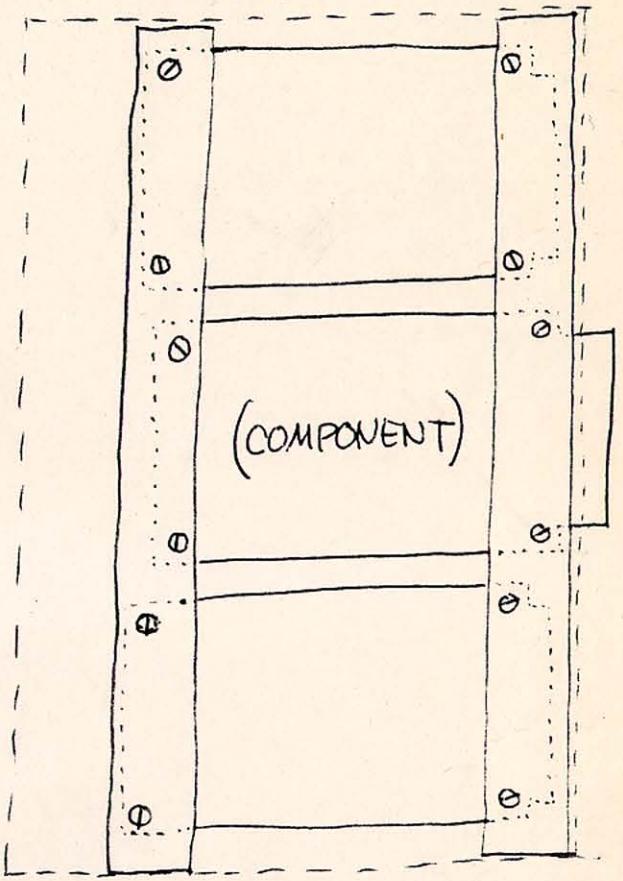


LEFT SIDE:



Crucial screw; circuits all run to ground (⏚) at this point.

RIGHT SIDE:



Star washer all 4-40 screws which hold boards to supports.

Use #6 x 3/8" panhead self thread - ing screws for mounting supports to chassis box (no washers).

ADDER MULTIPLIER:

The adder multiplier is used to add (superimpose), fade and gain control (multiply) signals.

JI1, JI2, JI3 and the inverted signal of JI7 are added together to form input channel A.

JI4, JI5, JI6 and the inverted signal of JI8 are added together to form input channel B.

The knobs above the connectors control the gain (contrast) of each individual input.

The amount of channel A and B mixed into the output, JO1 through JO4, is dependent on the position of R9 and the voltage inputted to JI9.
The effect of the knob position and the voltage are additive; the knob to the left and/or a maximum negative voltage on JI9 will cause channel B to be outputted only, similiarly, the knob to the right and/or a maximum positive voltage will cause channel A to be outputted only.
The knob at approximately the center with no voltage applied to JI9 will cause half-of channel A and half-of channel B to be added together and outputted.

TEST STUFF:

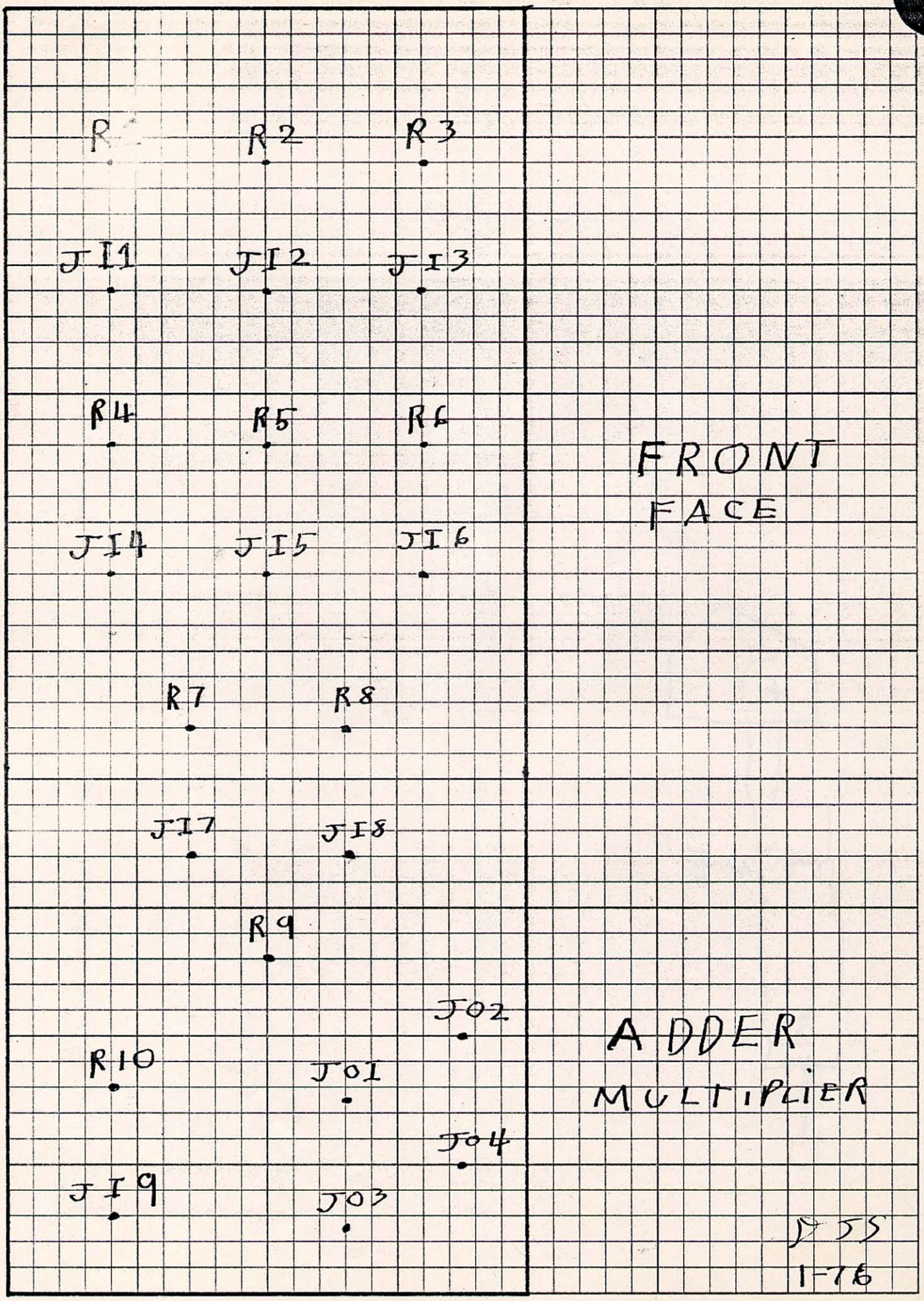
The adder multiplier should have a net gain of slightly greater than 1.
That is, a (+) or (-) .5 volt signal into the module should result in an undistorted output of approximately the same magnitude into a 75 ohm load.

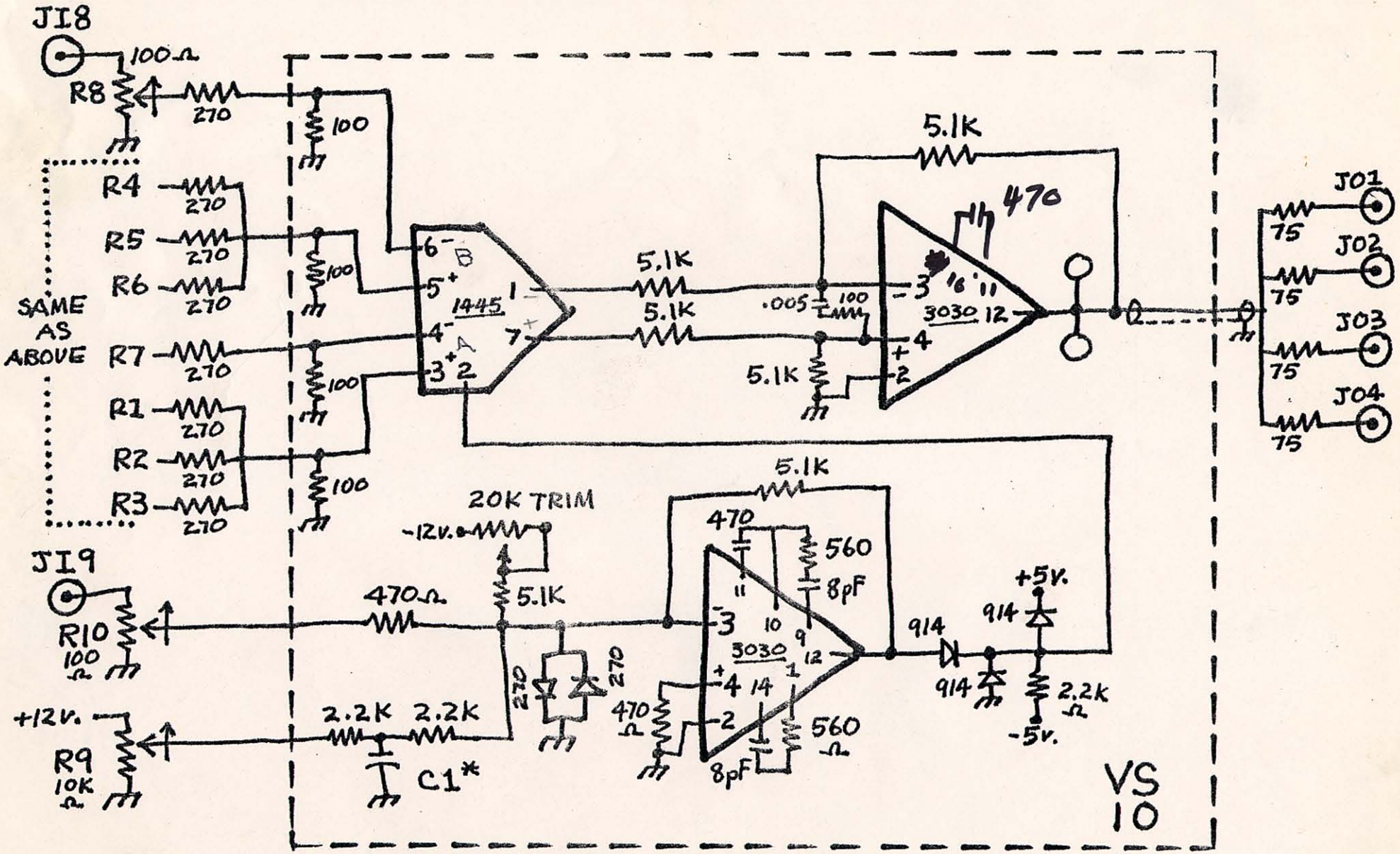
With no input the output should be approximately 0 volts (+ or - .05 volts).
Adjust 20k trimmer pot so with R9 in center position and no input to JI9 channel A and channel B have equal gain.

C1* STUFF:

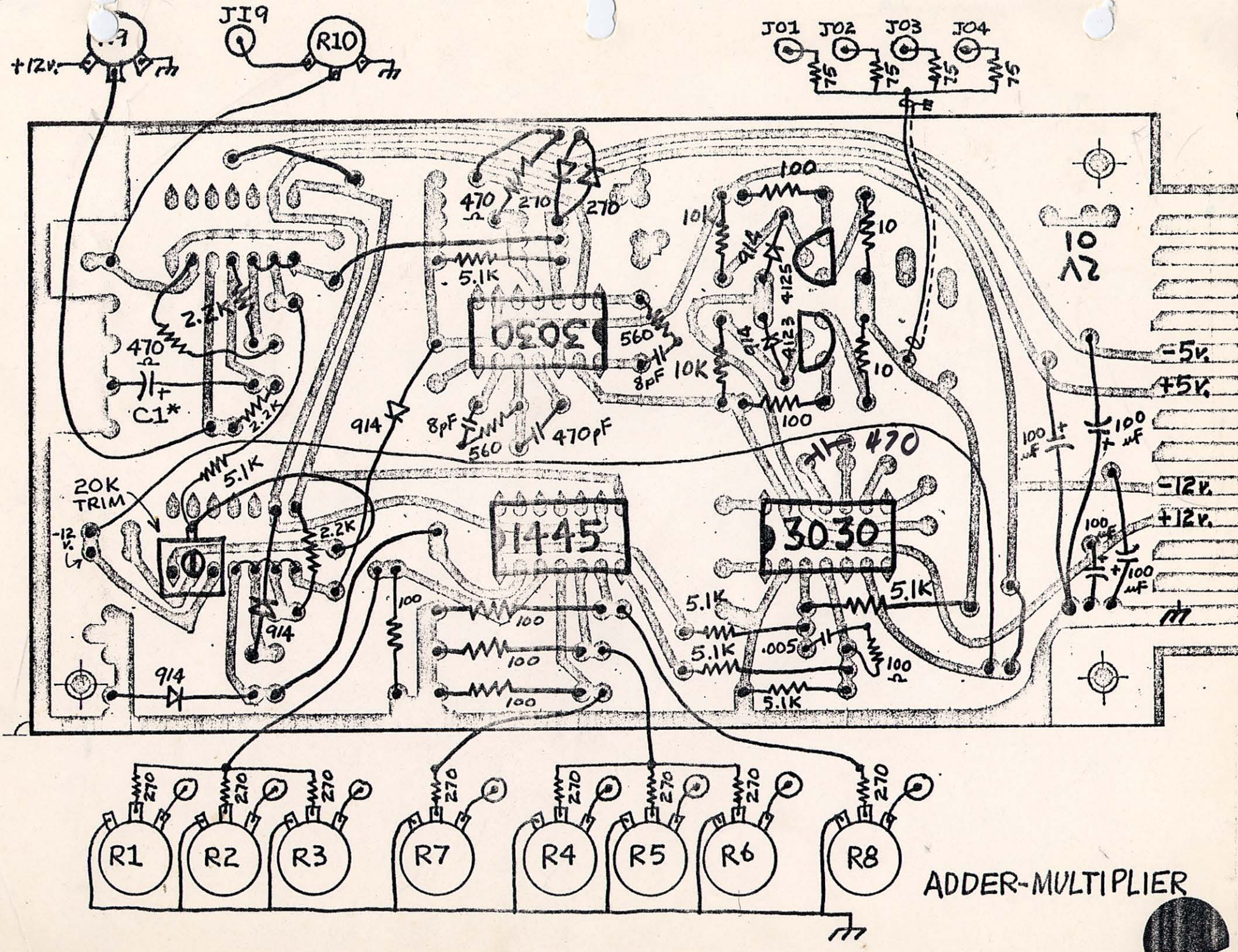
The capacitor, C1, is used to filter the bias control, R9. One may choose a value which will vary the 'feel' of the knob.

- 20uF is a minimum value which will remove some noise...
- 50uF is the minimum value that I use; it doesn't affect the feel of the knob...
- 100uF removes some shakiness of the hand (included in the parts list)...
- 500uF is Phil's recommendation (very slushy feeling)...

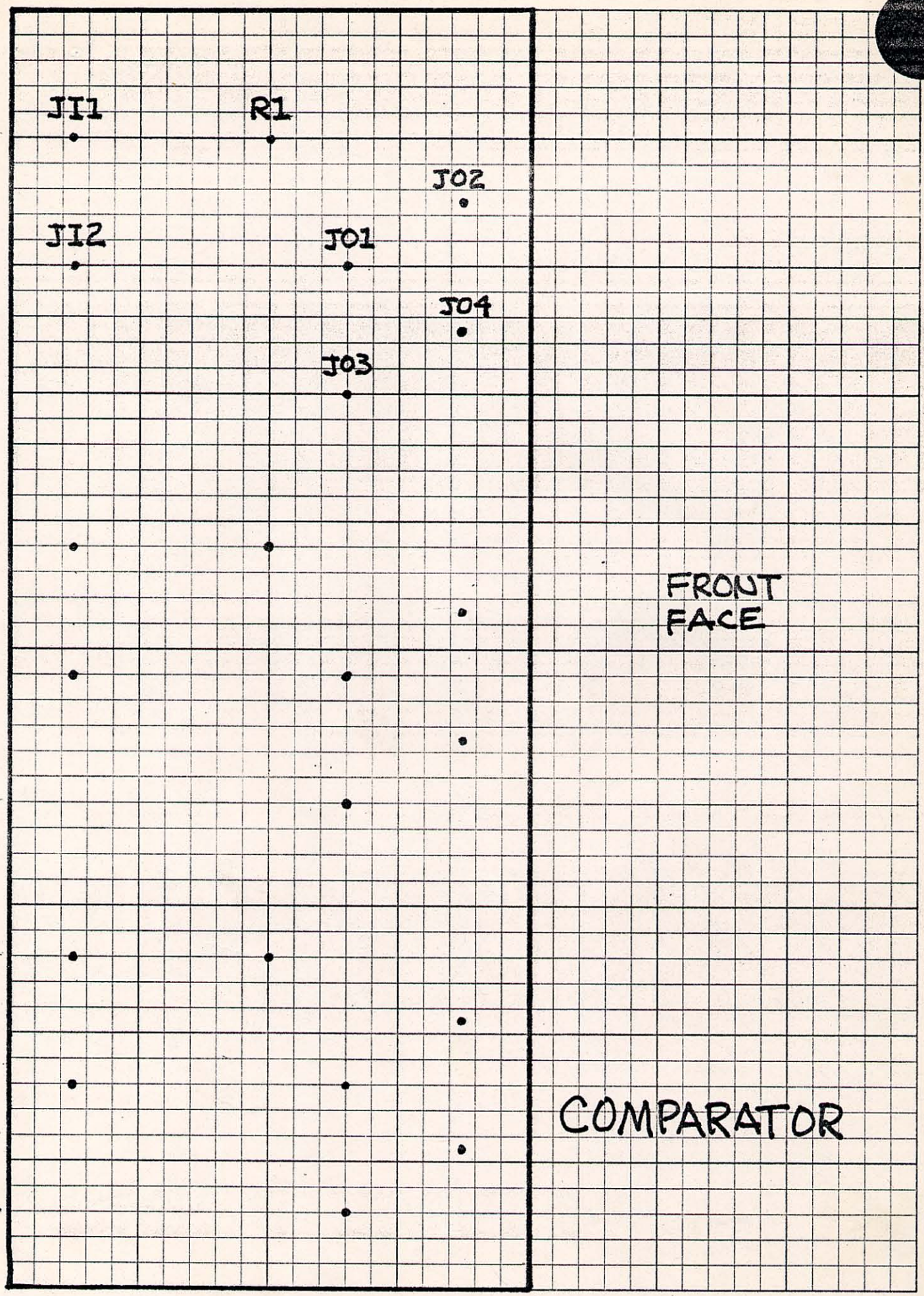

R1
R2
R3
JI1
JI2
JI3
R4
R5
R6
JI4
JI5
JI6
R7
R8
JI7
JI8
R9
JO2
R10
JO1
JO4
JI9
JO3
FRONT
FACE
ADDER
MULTIPLIER
S 55
1-76

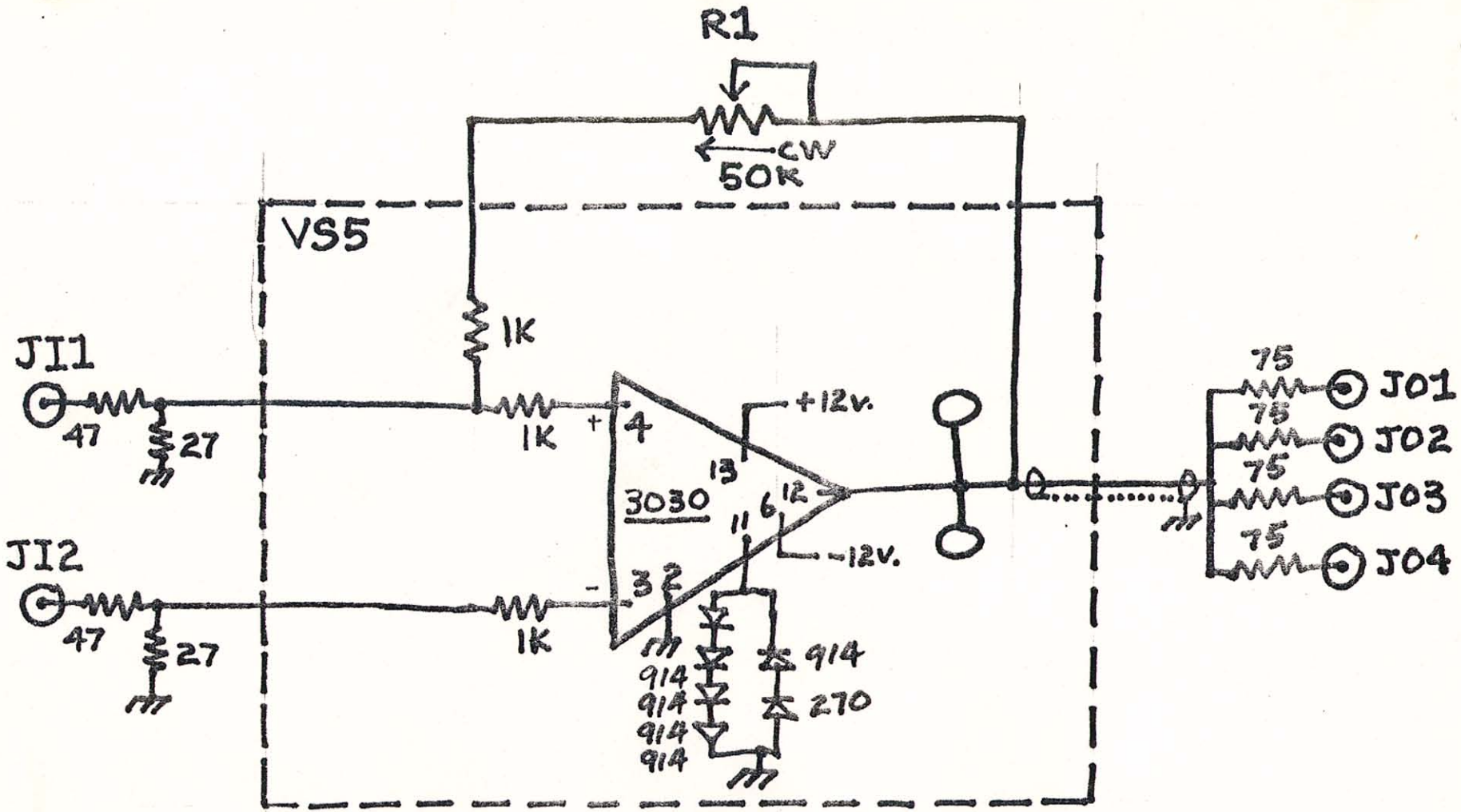


ADDER-MULTIPLIER

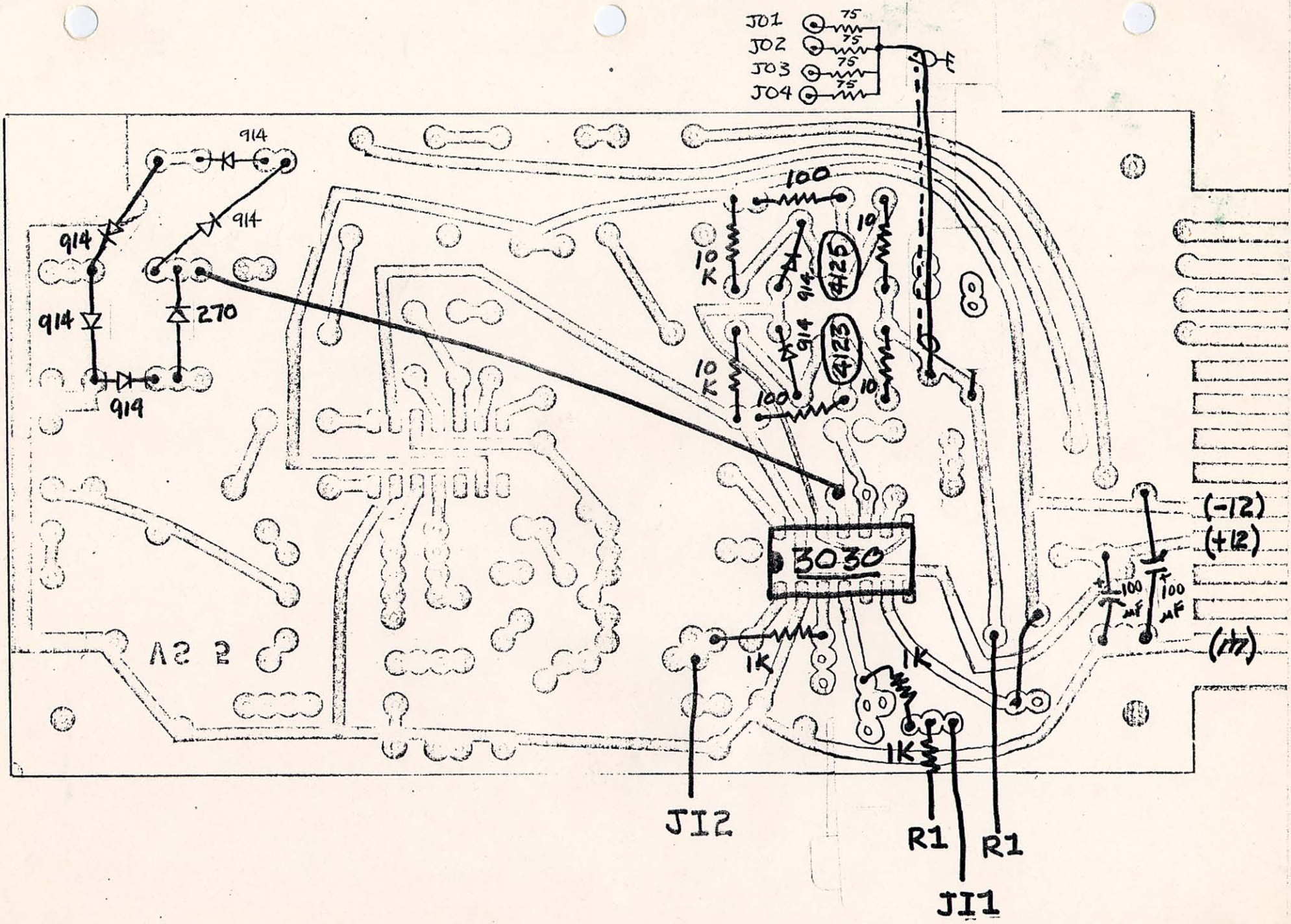

JI9
R10
+12v.
J01
J02
J03
J04
75
75
75
75
470
270
5.1K
2.2K
470
C1*
2.2K
914
8pF
560
470pF
560
8pF
10K
914
4125
914
4123
100
10K
10
10
10
100
470
100 uF
100 uF
100 uF
100 uF
-5v.
+5v.
-12v.
+12v.
20K
TRIM
-12 v.
5.1K
2.2K
914
914
100
100
100
100
5.1K
5.1K
.005
5.1K
5.1K
100
5.1K
270
270
270
270
270
270
270
270
R1
R2
R3
R7
R4
R5
R6
R8
ADDER-MULTIPLIER

2		525-1060	10 OHM, 1/4 W RES	.06	A	AM
4		525-1165	75 OHM, 1/4 W RES	.06	A	AM
6		525-1202	100 OHM, 1/4 W RES	.06	A	AM
[illegible]		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
6		525-1263	270 OHM, 1/4 W RES	.06	A	AM
[illegible]		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
1		525-1356	1KΩ OHM, 1/4 W RES	.06	A	AM
5		525-1467	3.1KOHM, 1/4 W RES	.06	A	AM
2		525-1522	10K OHM, 1/4 W RES	.06	A	AM
2		525-1302	470 OHM, 1/4 W RES	.06	A	AM
3		525-1405	2.2KOHM, 1/4 W RES	.06	A	AM
1		525-1550	15K OHM, 1/4 W RES	.06	A	AM
1		525-1603	27K OHM, 1/4 W RES	.06	A	AM
8	10F454	101UA	100 OHM POT 1/4SFAB	1.71	N	AM
1	9F089	U20	10K OHM POT 1/4SFT	1.45	N	AM
1	12F9800	3389P	20K TRIM BD-MT	.65	N	AM
2	14F592	UM15-417J	470 PF,DIP-MICA CAP	.32	N	AM
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2	14F1269	UM5-0800	8 PF,DIP-MICA CAP	.36	N	AM
6	710-1260	7-6-1000	100 MF,25VDC,ELEC.	.24	A	AM
1		.005				
5	553-0914	1N914B	SIL-DIODE	.19	A	AM
2		1N270	GERM-DIODE	.22	S	AM
1		2N4123	NPN TRANS	.22	S	AM
1		2N4125	PNP TRANS	.27	S	AM
2		CA3030	DIP OP-AMP,RCA	1.32	S	AM
1		MC1445L	DIP GAIN-CONT.AMP	1.90	S	AM
1		VSIO PC BOARD				AM
[illegible]		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
13	59F1337	13-236	BNC,FM-CHS.MT.	.72	N	AM
1		CHASSIS,AM-FACE		8.25	DG	AM
4	RB-67-1-DC-M-L-9	KNOB,BLACK/INLAY299		1.00	R	AM
4	RB-67-1-DC-M-L-9	KNOB,BLACK/INLAYSIL		1.00	R	AM
1	RB-67-1-DC-M-L-9	KNOB,BLACK/INLAY120		1.00	R	AM
1	RB-67-0-DC-M-D-9	KNOB,BLK/INLAY 299		1.00	R	AM


JI1
R1
JO2
JI2
JO1
JO4
JO3
FRONT FACE
COMPARATOR


R1
CW
50K
VS5
1K
JI1
47
27
1K
+
4
+12v.
13
3030
12
6
11
-12v.
JI2
47
27
1K
-
3
2
914
914
914
914
914
914
270
75
75
75
75
J01
J02
J03
J04

COMPARATOR



COMPARATOR 9-75

		525-1060	10 OHM, 1/4 W RES	.06	A	COM
7		525-1117	27 OHM, 1/4 W RES	.06	A	COM
5		525-1155	47 OHM, 1/4 W RES	.06	A	COM
5		525-1165	75 OHM, 1/4 W RES	.06	A	COM
6		525-1202	100 OHM, 1/4 W RES	.06	A	COM
9		525-1356	1K OHM, 1/4 W RES	.06	A	COM
6		525-1522	10K OHM, 1/4 W RES	.06	A	COM
3	10F473	503UA	50K OHM POT 1/4SFAB	2.21	N	COM
6	710-1260	7-6-1060	100 MF,25VDC,ELEC.	.24	A	COM
21	555-0914	1N914B	SIL-DIODE	.19	A	COM
3		1N270	GERM-DIODE	.22	S	COM
3		2N4123	NPN TRANS	.22	S	COM
3		2N4125	PNP TRANS	.27	S	COM
3		CA3030	DIP OP-AMP,RCA	1.32	S	COM
5			VS 5 P-C BOARD			COM
18	330-1537	15-256	BNC,PC-CHS.MT.	.72	N	COM
1			CHASSIS,COMP-FACE	8.25	OG	COM
3	RB-67-1-DC-M-L-9		KNOB,GRAY/LINE,9-BL	1.00	R	COM

FUNCTION GENERATOR

The function generator generates an output which is an arbitary function (with up to two points of inflection) of the input at J11. This results in an effect that is similiar to but more complex and controllable than photographic solerization.

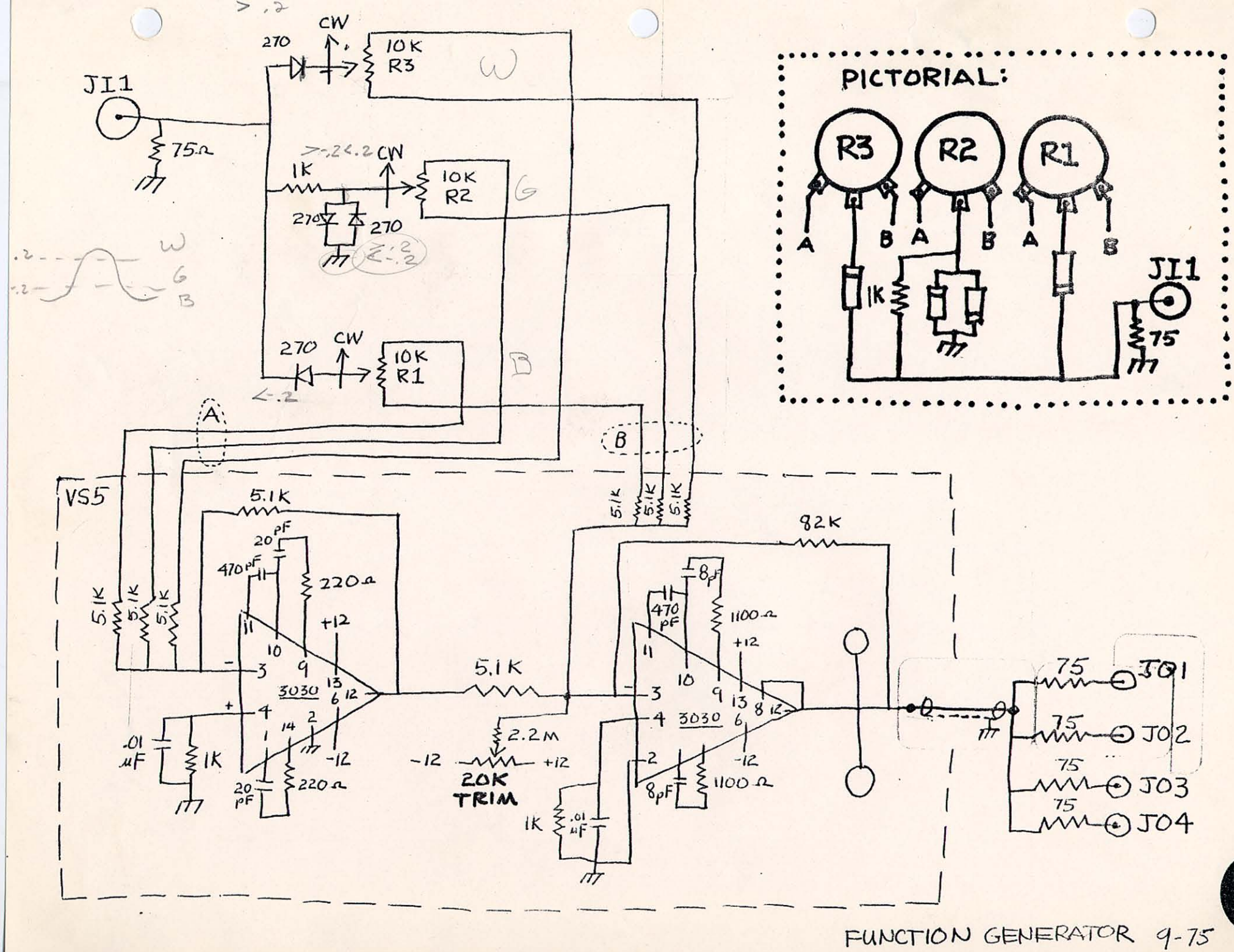
The function is controlled by R1, R2, and R3.
R1 controls the slope of the function for large negative inputs.
R2 controls the slope of the function for inputs near 0 voltages.
R3 controls the slope of the function for inputs of large positive voltage.

Clockwise is positive slope; counterclockwise is negative slope.

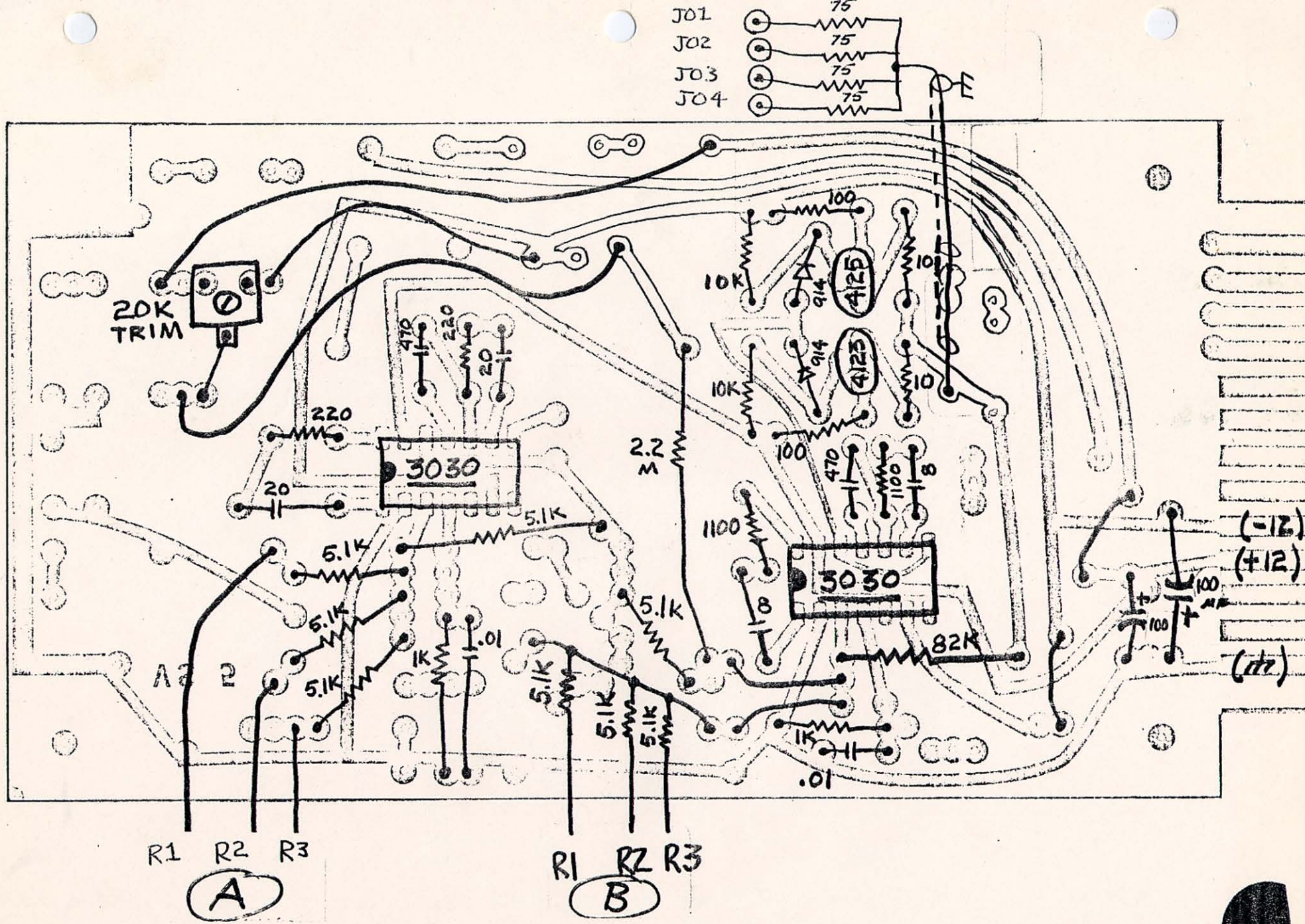
There are three electrical modules in one chassis box, so replicate work three times. Remember to buss (connect) +12 and -12 and ground wires from middle board to top and bottom board. Soldering directly to the foil is convenient.

TEST STUFF:

The 20K trimming resister on the VS5 board is adjusted such that no input results in 0 output voltage + or - .05 volts.



FUNCTION GENERATOR 9-75



FUNCTION GENERATOR 9-75

	525-1060	10 OHM, 1/4 W RES	.06	A	FGN
	525-1105	75 OHM, 1/4 W RES	.06	A	FGN
	525-1202	100 OHM, 1/4 W RES	.06	A	FGN
	525-1248	220 OHM, 1/4 W RES	.06	A	FGN
	525-1356	1K0 OHM, 1/4 W RES	.06	A	FGN
	525-1361	1.1KOHM, 1/4 W RES	.06	A	FGN
	525-1467	5.1KOHM, 1/4 W RES	.06	A	FGN
	525-1522	10KKOHM, 1/4 W RES	.06	A	FGN
	525-1703	82KKOHM, 1/4 W RES	.06	A	FGN
	525-1907	2.2MOHM, 1/4 W RES	.06	A	FGN
10F498	1050A	10K OHM POT 1/4SFAB	2.21	N	FGN
12F9500	3389P	20K TRIM 80-MT	.65	N	FGN
14F1269	DM5-0800	8 PF,DIP-MICA CAP.	.36	N	FGN
14F557	DM15-2000J	20 PF,DIP-MICA CAP.	.18	N	FGN
14F542	DM15-417J	470 PF,DIP-MICA CAP.	.32	N	FGN
27F507	TDC-103Z	.01 MF,DSC-CER CAP.	.05	N	FGN
710-1200	7-G-1000	100 MF,25VDC,ELEC.	.24	A	FGN
555-0914	1N9148	SIL-DIODE	.19	A	FGN
	1N270	GERM-DIODE			FGN
	2N4123	NPN TRANS	.22	S	FGN
	2N4125	PNP TRANS	.27	S	FGN
	CA3030	DIP OP-AMP,RCA	1.32	S	FGN
		VS 5 P-C BOARD			FGN
39F1537	13-236	BNC,PM-CHS.MT.	.72	N	FGN
		CHASSIS,F.G. FACE	8.25	DG	FGN
RB-67-1-DC-M-L-9		KNOB;#299,MAT-BLACK			
		INLAY ,IND.7WHITE			
		-9 /WHITE	1.00	R	FGN

DIFFERENTIATOR

The differentiator produces an output which is proportional to the rate of change of the input signal. Fast rates of change correspond to edges in a picture and are preferentially amplified by the module.

JI6 amplifies only the sharpest edges...
JI5 amplifies the sharpest edges and slightly softer edges...
JI4, JI3 and JI2 amplify progressively softer and softer edges until by JI1 almost all of the whole picture is amplified.

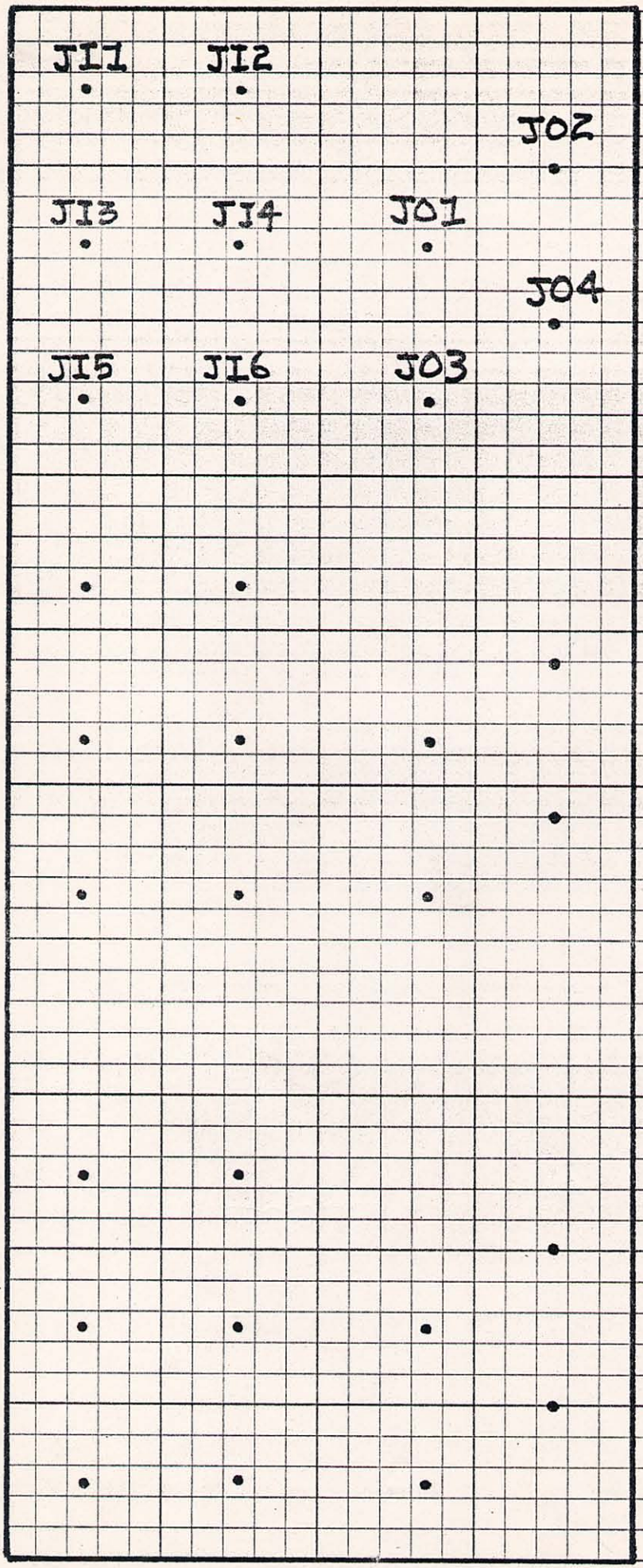
There are three electrical modules in one chassis box. One diagram is supplied, so replicate work three times. Remember to buss (connect) +12, -12 and ground from the center board to the upper and lower boards; soldering directly to the foil or connecting corresponding bypass capacitors is convenient.

TEST STUFF:

The module should amplify high frequency (greater than 20 kHz) sine waves with greater gain than lower frequency sine waves. The sine waves should be undistorted.

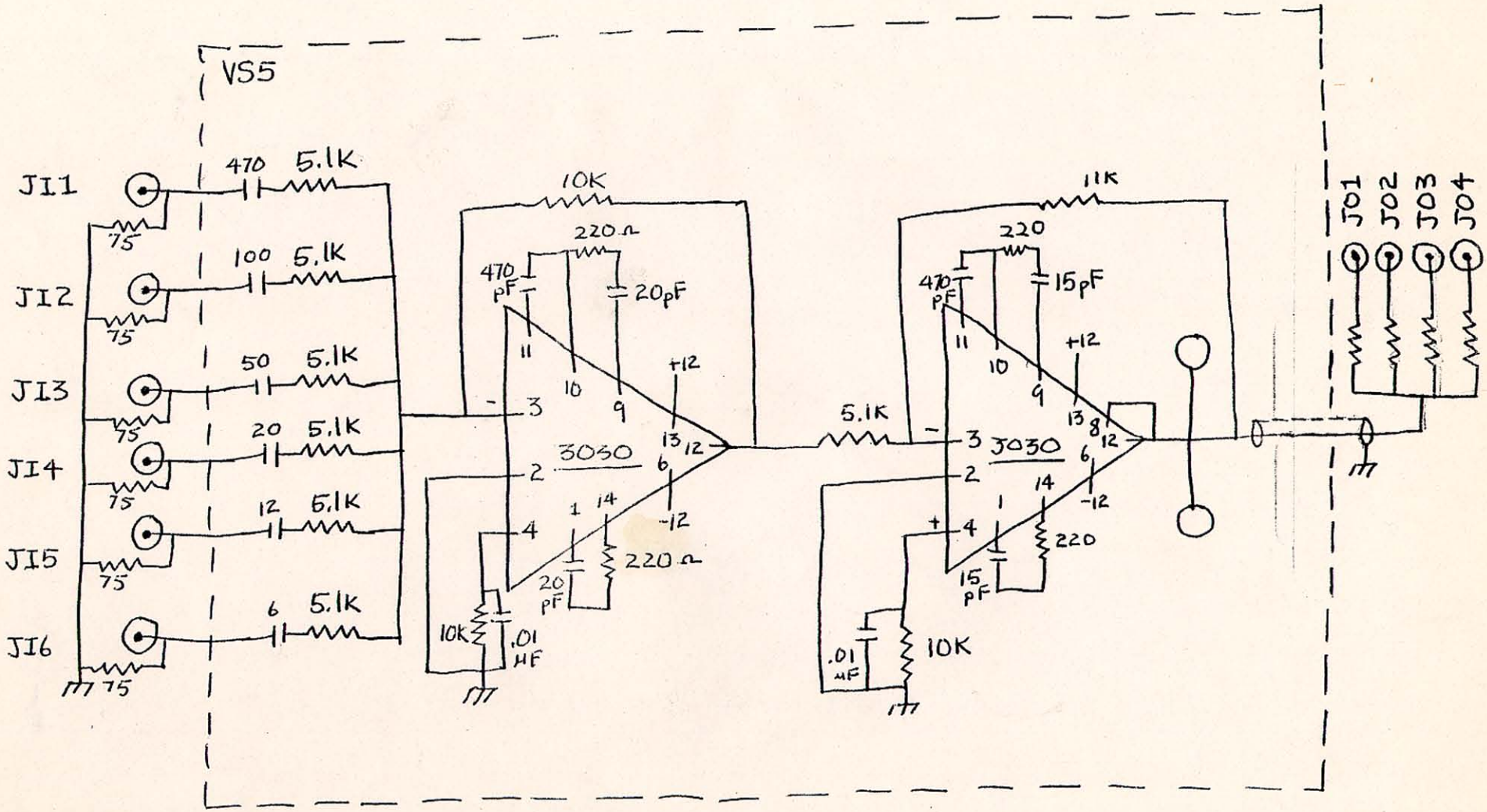
Square waves should be differentiated; that is, there should be a positive spike associated with the rising edge of the square wave, and a negative spike associated with the falling edge of the square wave.

No input should result in 0 volts output + or - .05 volts.

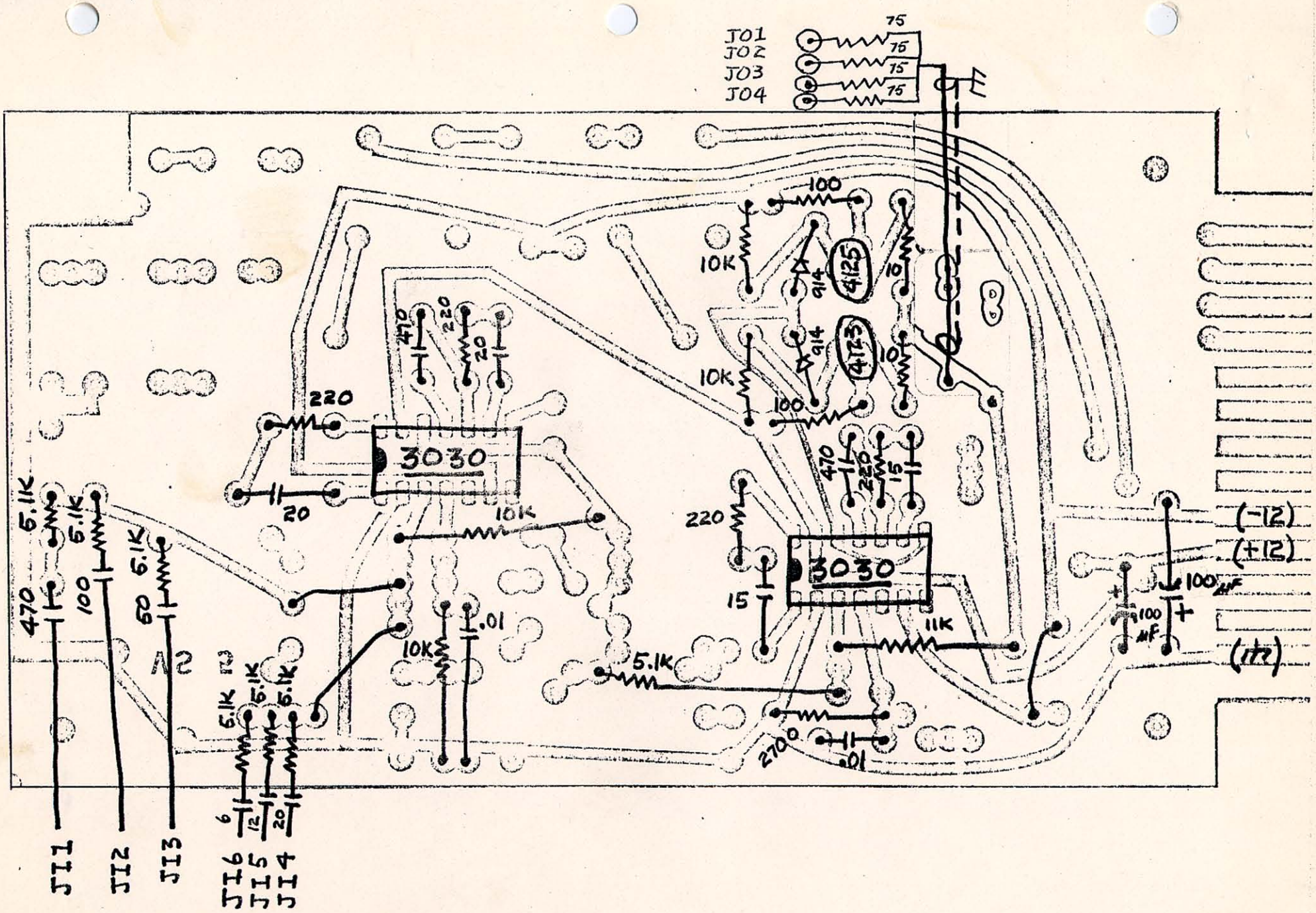


FRONT FACE

DIFFERENTIATOR



DIFFERENTIATOR 9-15



DIFFERENTIATOR 9-15

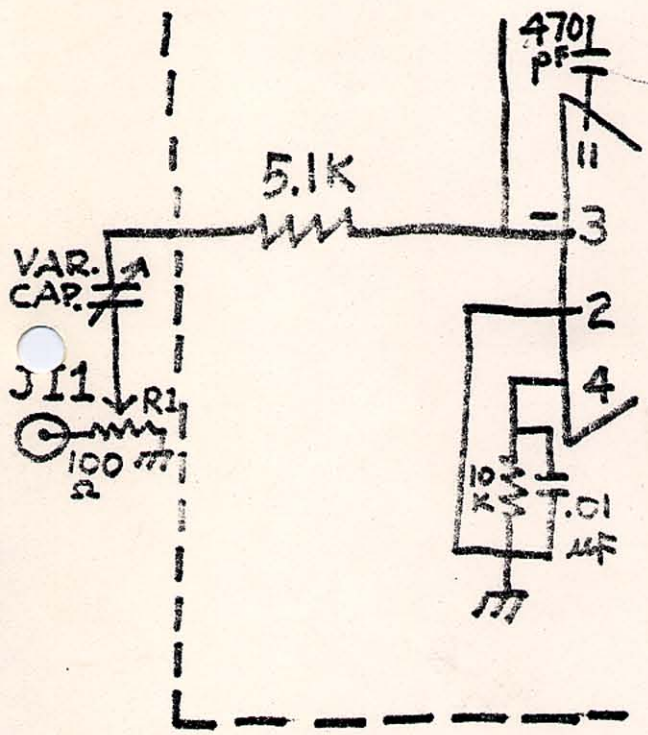
ADDENDUM

This addendum provides brief data for an optional differentiator (opt. diff.). The opt. diff. has some trade-offs compared to the original differentiator (orig. diff.). Consider the following and evaluate for yourself:

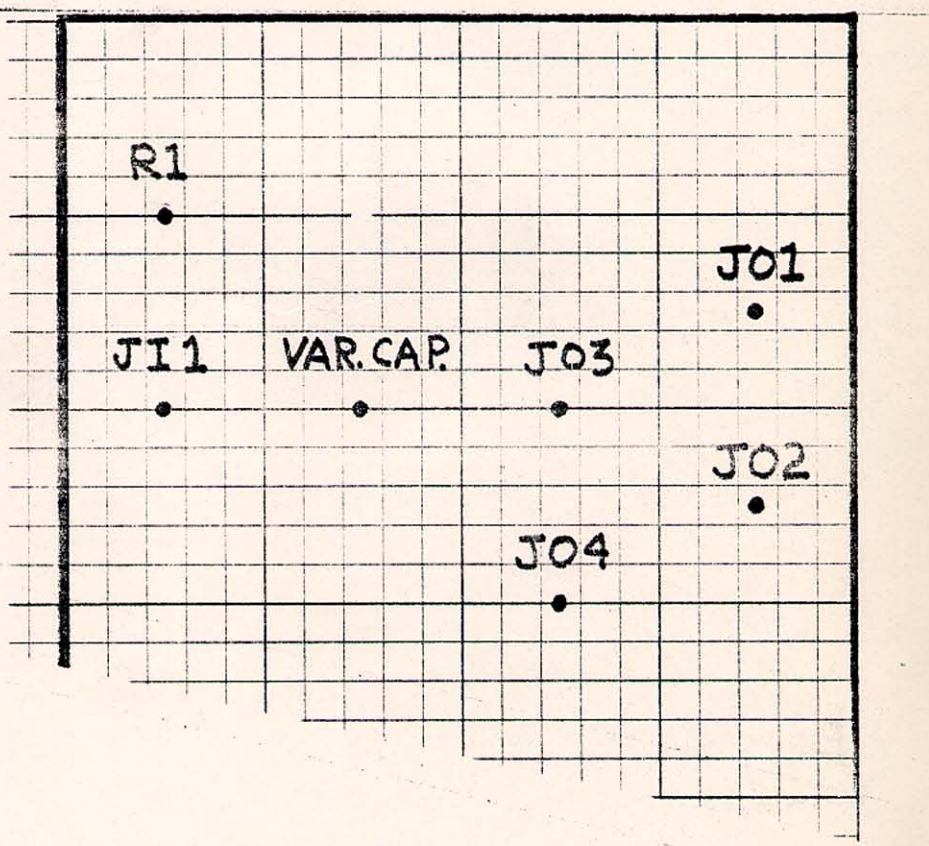
In the orig. diff. you input a signal via any 6 BNC inputs (JI1-thru-JI6); and, in the opt. diff. you input a signal to 1 BNC input (JI1), control its gain with R1, and its differentiation constant via a variable capacitor (VAR. CAP.). The VAR. CAP. will give you the same approximate differentiation constants as JI2-thru-JI6 in the orig. diff.; but, will <u>not</u> give you the largest differentiation constant available at JI1 in the orig. diff.

PART NUMBER FOR THE VAR. CAP. IS: <u>ALLIED #695-2300 (7.2pf-151pf) $9.00/ea.</u>

<u>SCHEMATIC</u> for opt. diff. -



<u>FRONT FACE</u> for opt. diff. -



		525-1060	10 OHM, 1/4 W RES	.06	A	DIF
24		525-1165	75 OHM, 1/4 W RES	.06	A	DIF
9		525-1202	100 OHM, 1/4 W RES	.06	A	DIF
12		525-1248	220 OHM, 1/4 W RES	.06	A	DIF
21		525-1467	5.1KOHM, 1/4 W RES	.06	A	DIF
15		525-1522	10KKOHM, 1/4 W RES	.06	A	DIF
3		525-1528	11KKOHM, 1/4 W RES	.06	A	DIF
3	14F1267	DM5-0600	6 PF,DIP-MICA CAP	.36	N	DIF
3	14F554	DM15-120J	12 PF,DIP-MICA CAP	.20	N	DIF
6	14F555	DM15-150J	15 PF,DIP-MICA CAP	.20	N	DIF
9	14F557	DM15-200J	20 PF,DIP-MICA CAP	.18	N	DIF
3	14F567	DM15-500J	50 PF,DIP-MICA CAP	.19	N	DIF
3	14F557	DM15-101J	100 PF,DIP-MICA CAP	.18	N	DIF
9	14F592	DM15-471J	470 PF,DIP-MICA CAP	.32	N	DIF
6	47F307	TDC-103Z	.01 MF,DSC-CER CAP.	.05	N	DIF
3	719-1200	7-3-1000	100 MF,25VDC,ELEC.	.24	A	DIF
6	353-2914	1N9146	SIL-DIODE	.19	A	DIF
3		2N4123	NPN TRANS	.22	S	DIF
3		2N4125	PNP TRANS	.27	S	DIF
5		CA3030	DIP OP-AMP,RCA	1.32	S	DIF
30	59F1337	13-236	BNC,FM-CHS.MT.	.72	N	DIF
3			VS 5 P-C BOARD			DIF
1			CHASSIS,DIF-FACE	8.25	DG	DIF

3	(see ADDENDUM)	695-2300	7.2-151 pf, var. cap.	9.00	A	DIF

SYNC STRIPPER and CAMERA INPUT

This module performs several related utility functions.

A video signal is inputted to JI1; this signal is clamped and sync suppressed and is available at JO1, JO2, JO3 and JO4. This part is identical to one-third of the INPUT module except the composite sync is generated internally (consult INPUT module documentation for explanation).

In addition, the video signal inputted at JI1 is separated from the sync information by the sync strip card. The vertical sync is filtered and amplified by the vertical filter-amp and distributed to output jacks.

Similiarly, the horizontal sync information is filtered and amplified and distributed to output jacks.

Burst flag and blanking information is regenerated from the horizontal and vertical sync and distributed to output jacks.

Vertical sync (-4v.) is available at JO13, JO14 and at pin#2 of the EIAJ (6-pin) camera connectors.

Horizontal sync (-4v.) is available at JO9, JO10 and at pin#5 of the EIAJ (6-pin) camera connectors.

Blanking is available at JO11 and JO12. Burst is available at JO5 and JO6

The video signal (from the camera) with composite sync is made available at the BNC connector above the corresponding EIAJ (6-pin) camera connectors.

When this module is used, the sync for the IP is stripped from the video signal inputted to JI1. If a camera is used for this purpose it should of course not be be sunk to the IP; but must be internally sunk or sunk from a non-IP source.

TEST STUFF:

R2 and R3 should be adjusted the same as R1 and R2 in the INPUT module.
The trimmer on the vertical filter amp should be adjusted so the vertical signal out is the same length as the vertical sync present in the original signal.

The trimmer on the horizontal filter amp should be adjusted so the horizontal signal out is the same as the horizontal sync in the original signal. (NOTE: these adjustments are hard to make, but are not very critical in timing).

R4, front panel associated with the sync stripper, should be adjusted to minimize any jitter in output picture.

The blanking and burst amp is a set of three identical circuits except for the timing capacitors. Referring to the schematic diagram, the first half of the 9602 sets a delay time to the pulse and the second half times the pulse.

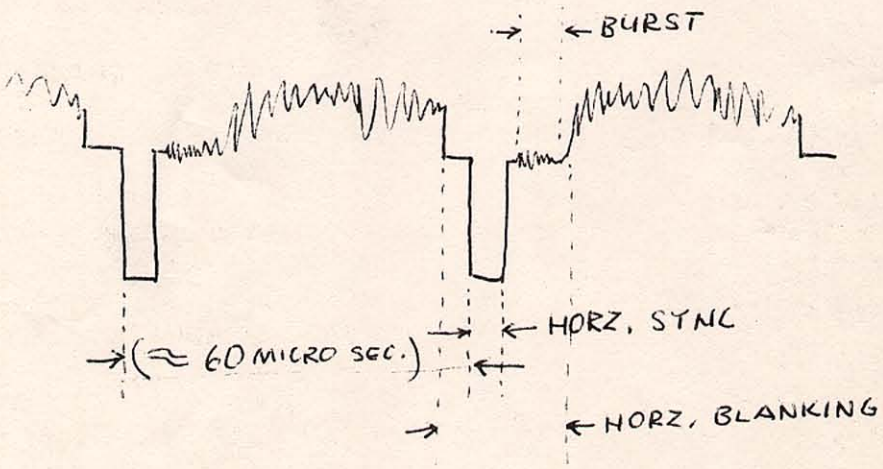
In the case of the burst flag R1T sets the delay from the beginning of the horizontal sync pulse to the beginning of the burst flag, and R2T determines the length of the burst flag.

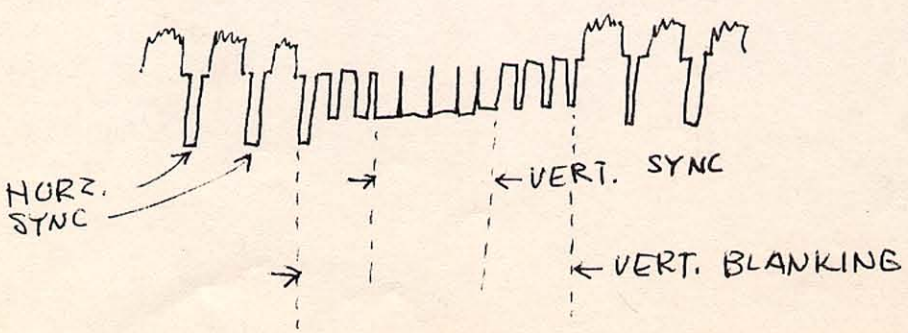
In blanking, R3T sets the delay from the beginning of the horizontal sync pulse to the beginning of the blanking pulse for the next horizontal line. This period is slightly less than on horizontal line. R4T sets the length of the blanking pulse.

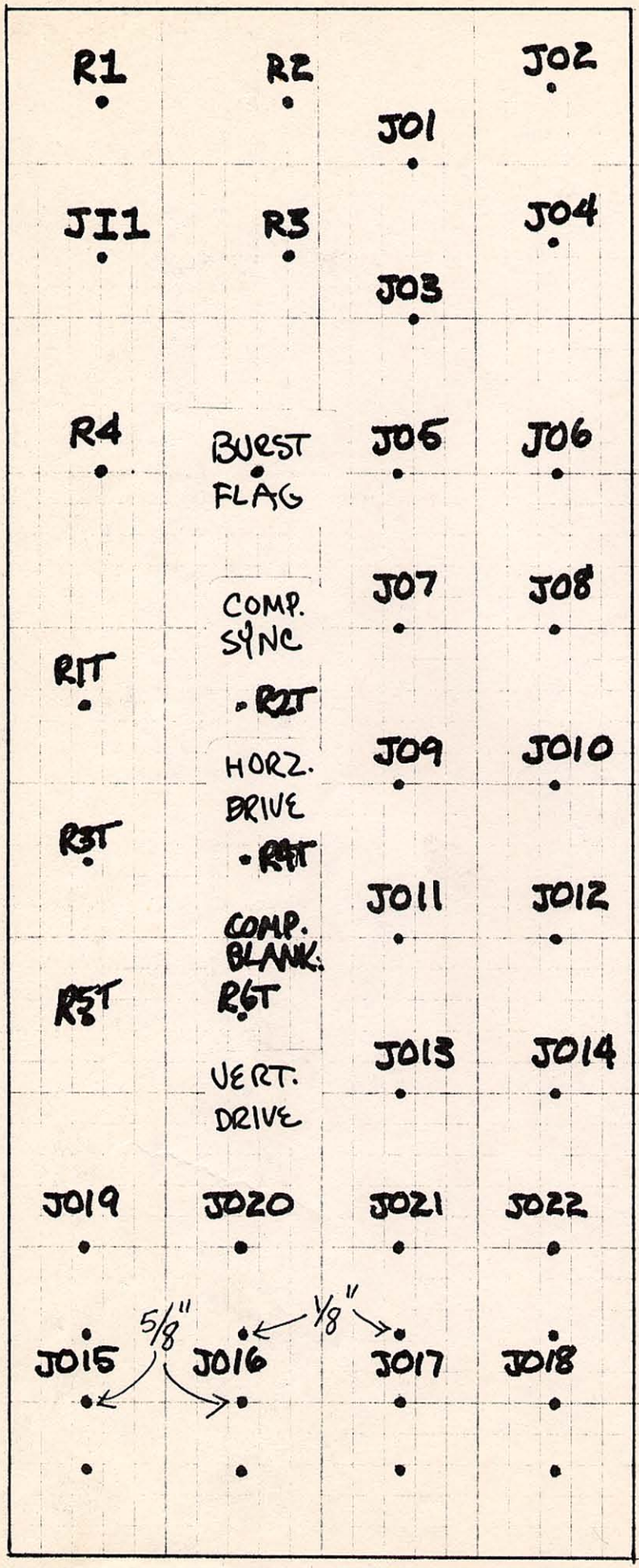
Vertical blanking is similiar with R5T setting the delay from the beginning of the vertical sync pulse to the beginning of the vertical blanking interval. R6T sets the length of the blanking interval.

To adjust all of these, feed into the module a high quality video signal (from a clearly received broadcast station or from the color encoder in the IP driven by a high quality sync generator). Adjust the output pulses from the sync strip to be identical with the pulses from the standard source.

HINT: start with all pots turned nearly full clockwise (minimum resistance). If the resistance is too high the device stays on all the time and if the resistance is too small, pulse may be too short to be seen on an inexpensive osscilloscope. A dual-trace triggered osscilloscope is preferred but a single trace scope can be used.



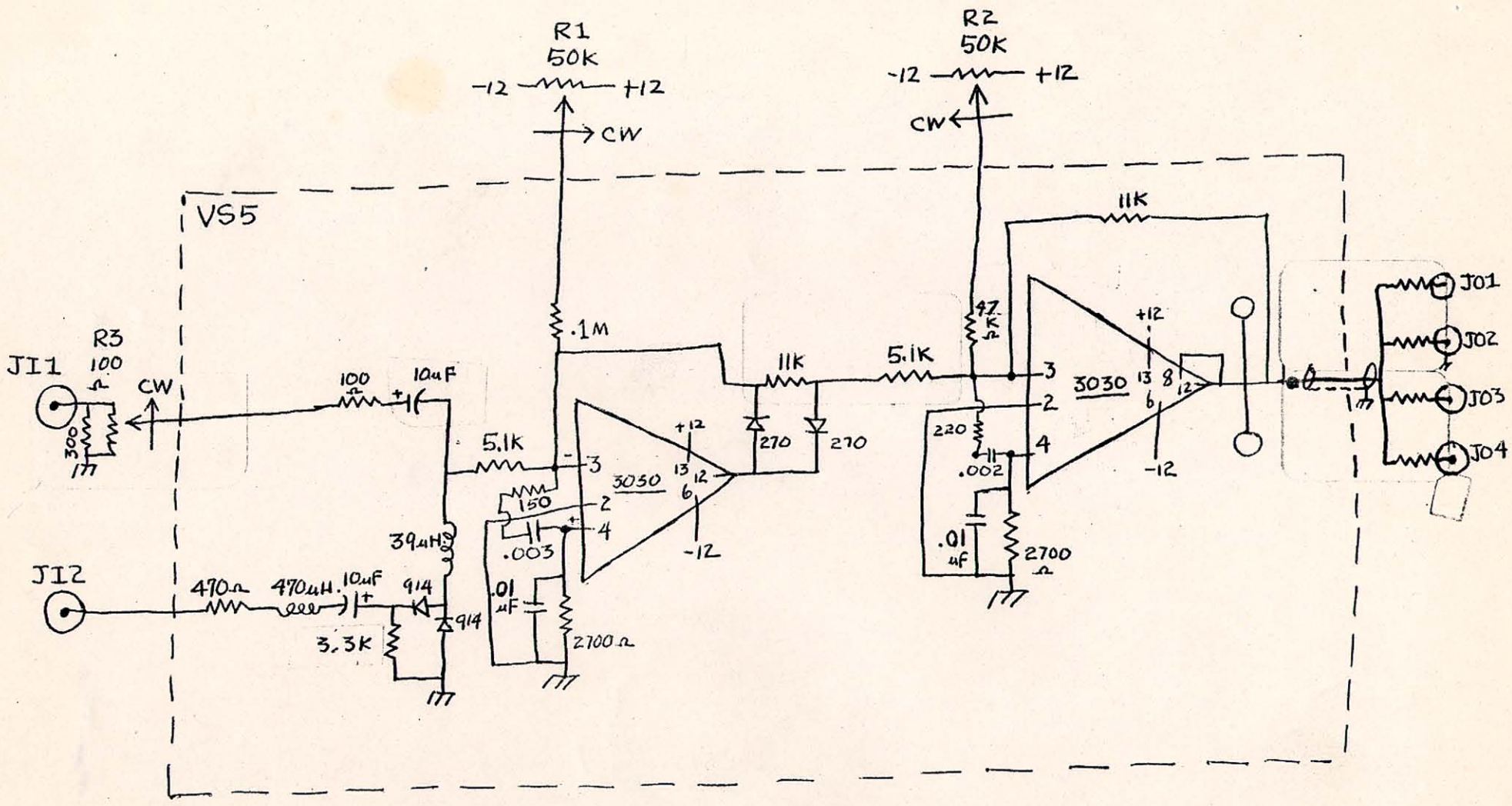


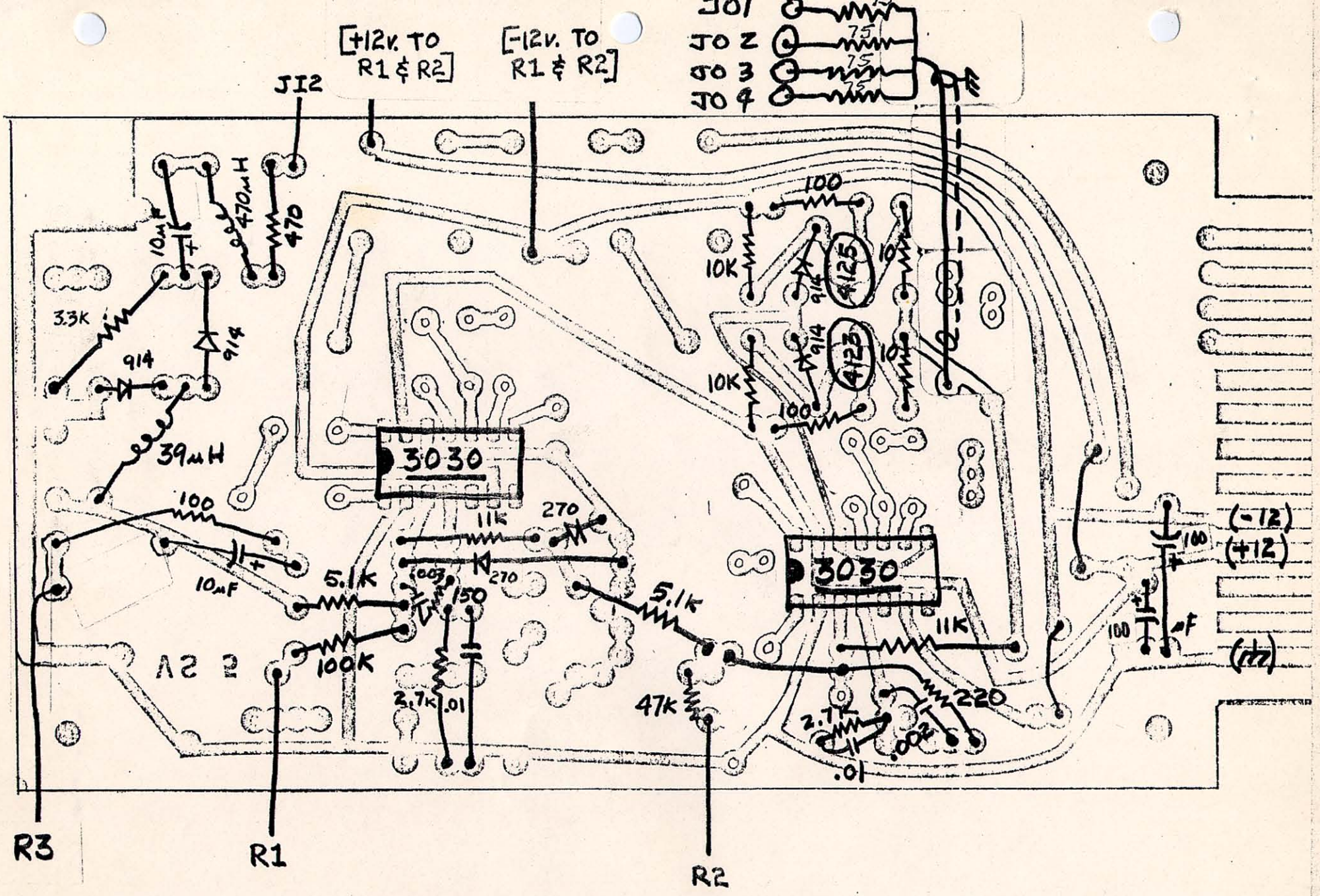


OK 9-75

FRONT FACE

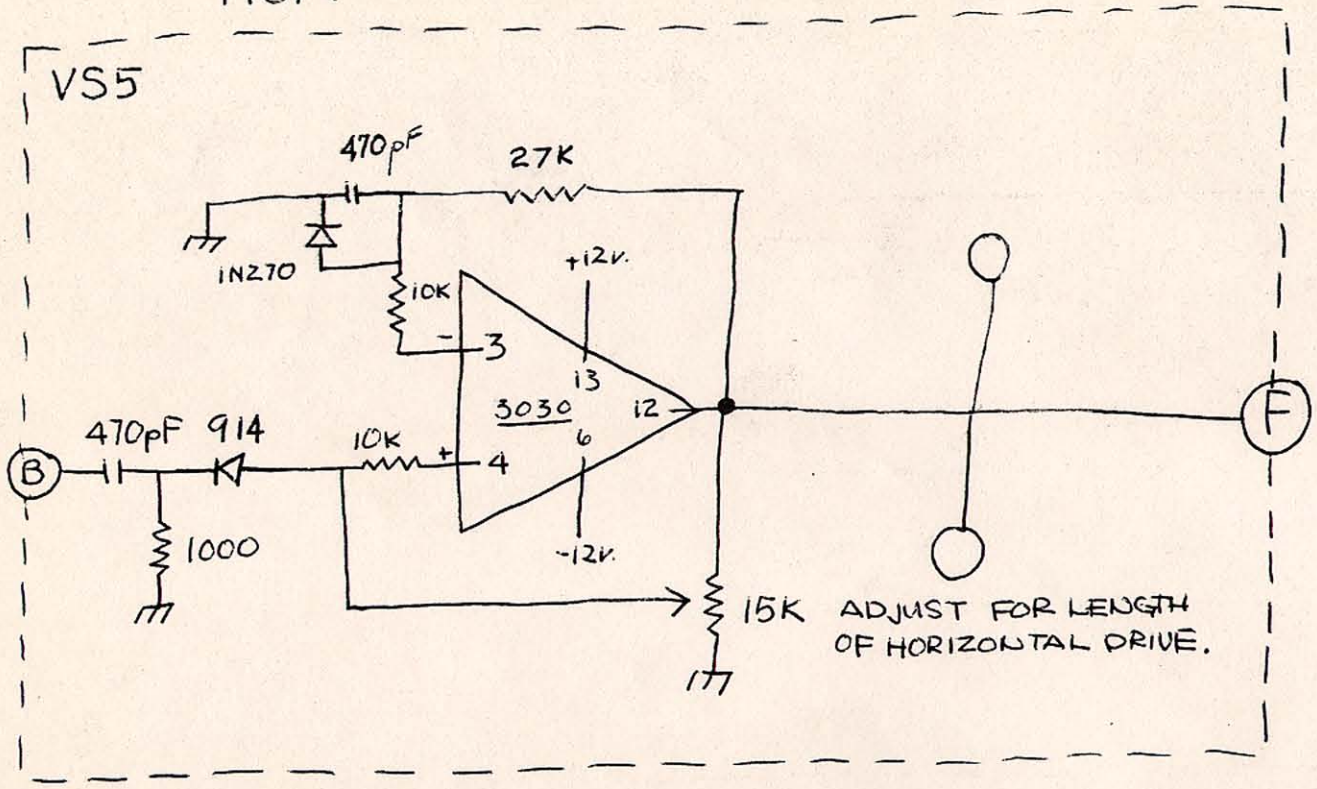
SYNC STRIPPER & CAMERA INPUT


R1
50K
-12
+12
CW
R2
50K
-12
+12
CW
VS5
JI1
R3
100
CW
300
100
10uF
.1M
11K
5.1K
47K
3030
+12
-12
JO1
JO2
JO3
JO4
5.1K
150
.003
3050
270
270
220
.002
.01 uF
2700
JI2
470
470uH
10uF
914
914
3.3K
39uH
.01 uF
2700
INPUT


[+12V. TO R1 & R2]
[-12V. TO R1 & R2]
JO1
JO2
JO3
JO4
75
75
75
JI2
10mF
470mH
470
3.3K
914
914
39mH
100
10mF
5.1K
.003
270
11K
270
150
100K
3.7K
.01
47K
5.1K
3030
3030
10K
914
4125
10
100
10K
914
4123
10
100
11K
220
2.7K
.002
.01
100
mF
100
(-12)
(+12)
A2 2
R3
R1
R2

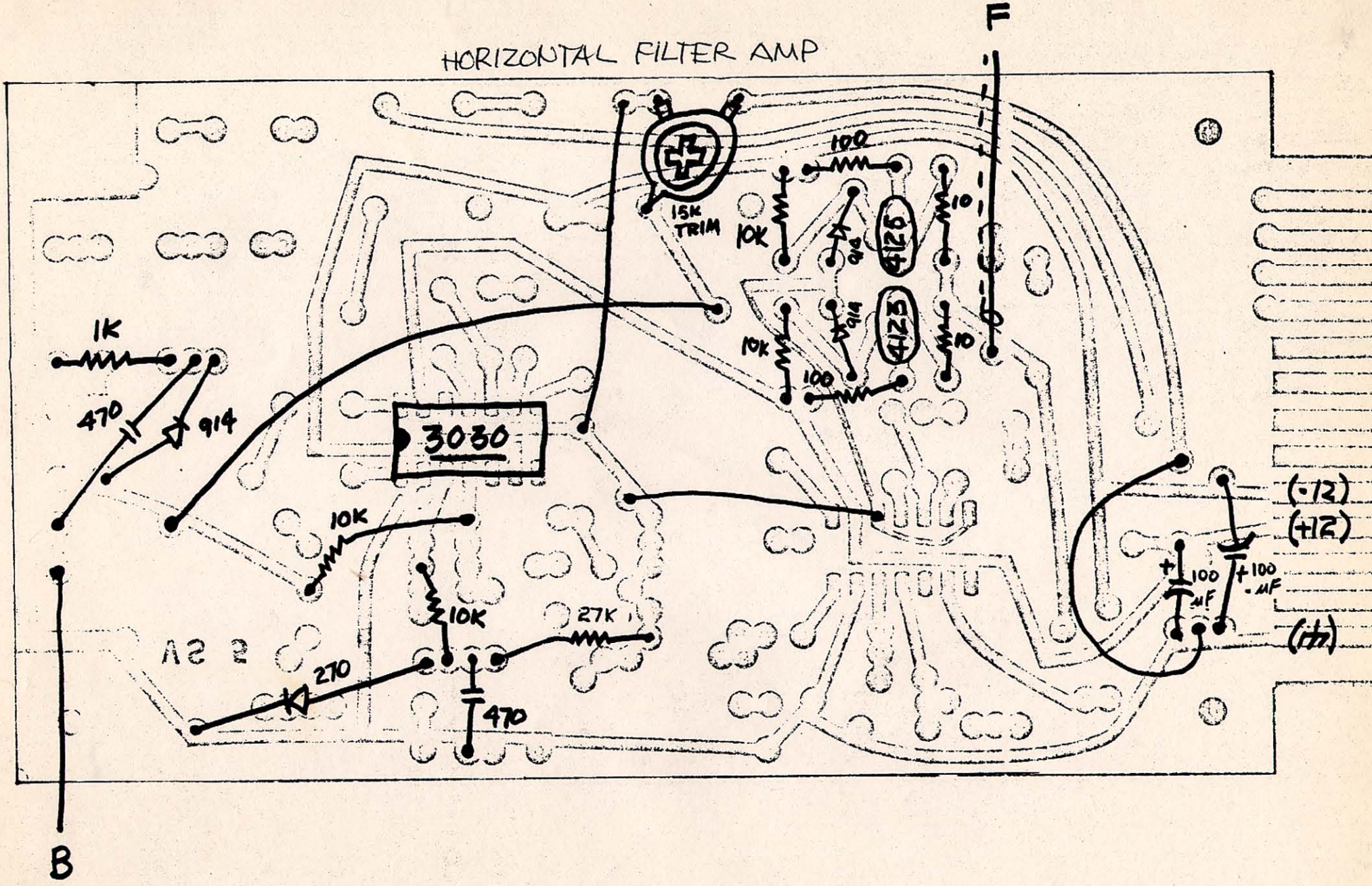
INPUT 9-75

HORIZONTAL FILTER AMP



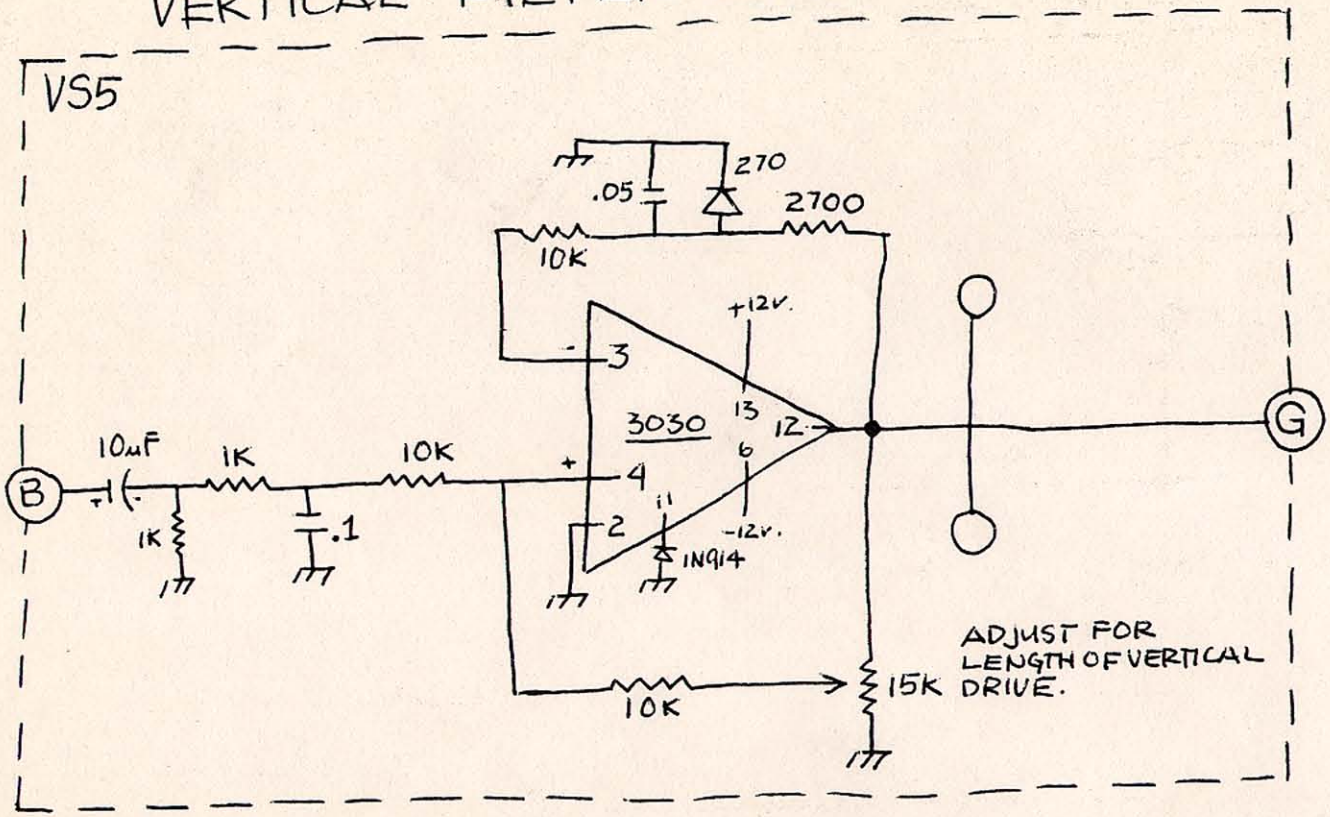
SYNC STRIPPER & CAMERA INPUT

HORIZONTAL FILTER AMP



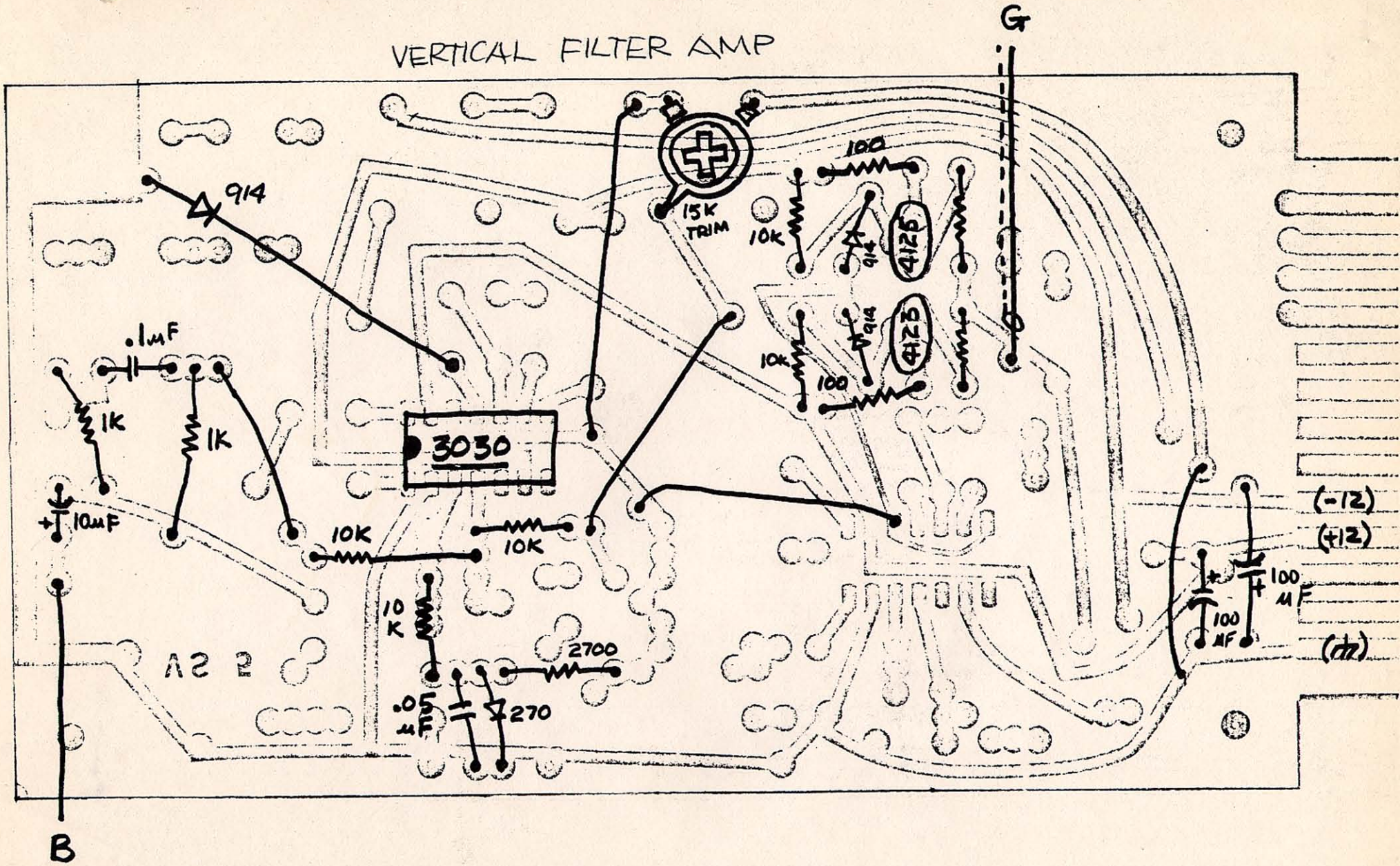
SYNC STRIPPER & CAMERA INPUT

VERTICAL FILTER AMP



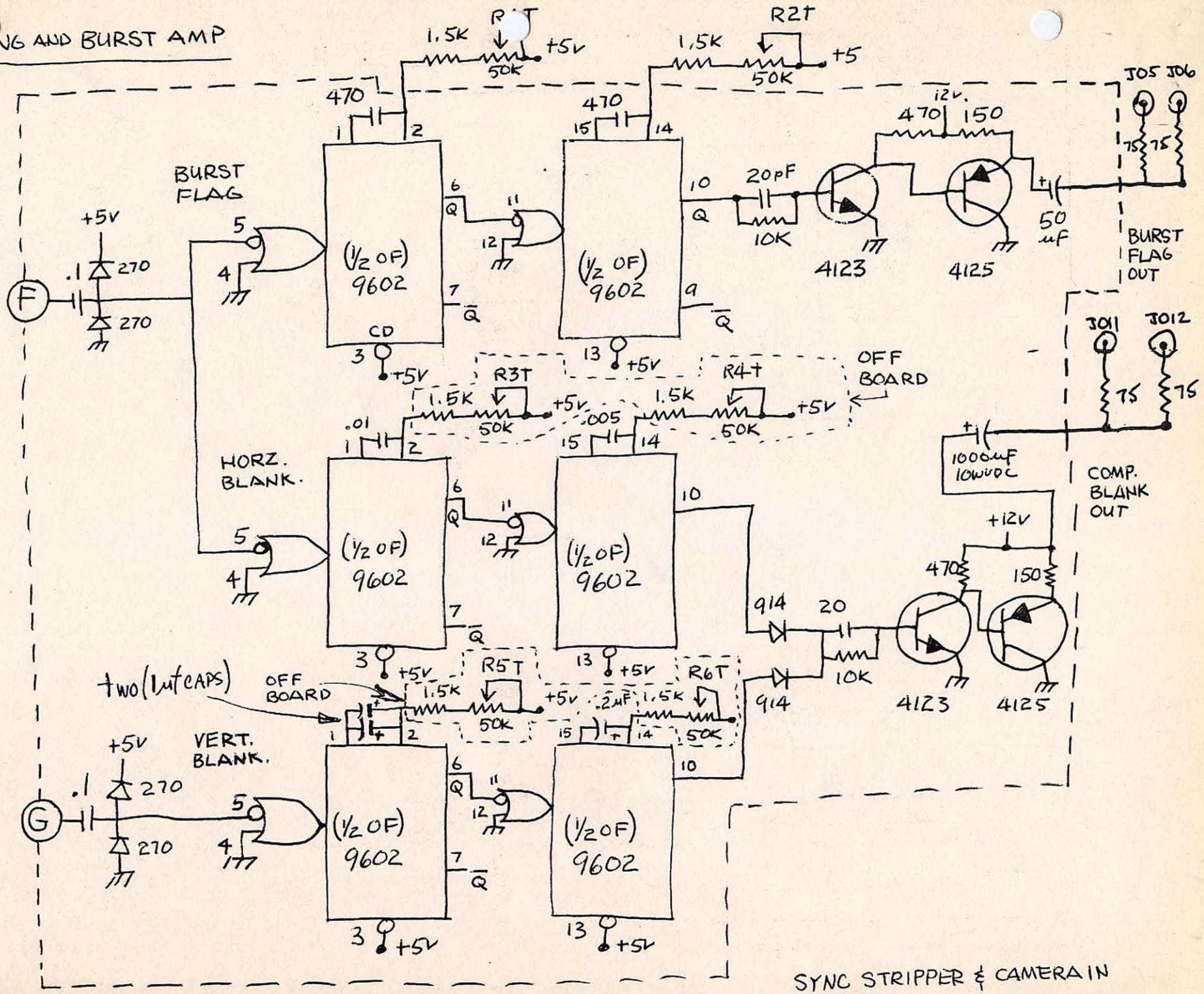
SYNC STRIPPER & CAMERA INPUT

VERTICAL FILTER AMP



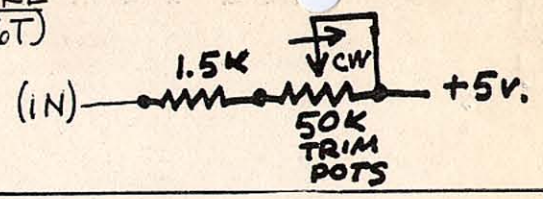
SYNC STRIPPER & CAMERA INPUT

BLANKING AND BURST AMP

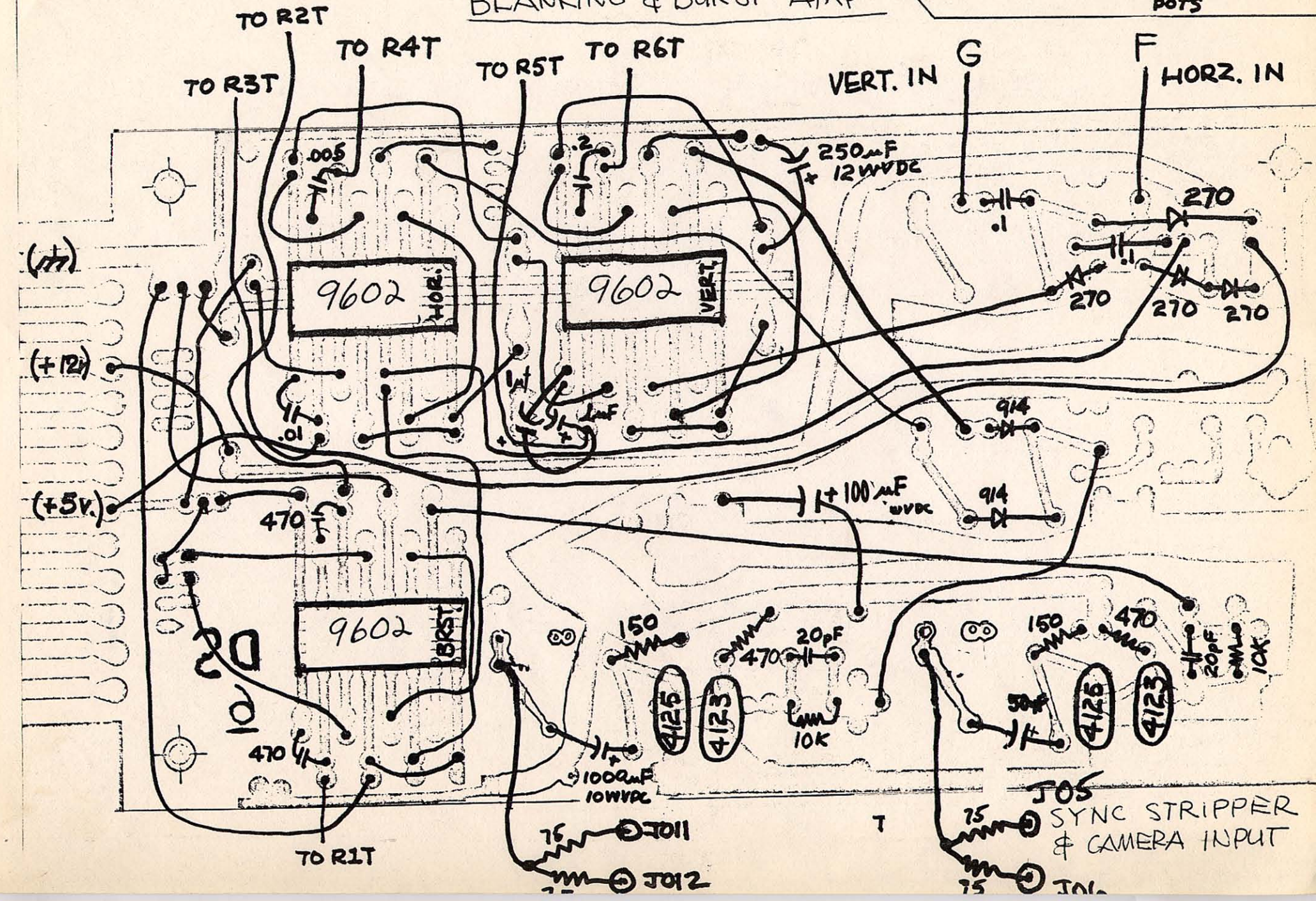


R1T THROUGH R6T (TRIM POTS) SHOULD BE MOUNTED ON A HOME-MADE PLASTIC, MASONITE??? BOARD PUT IN THE LEFT SIDE (FROM FRONT VIEW) OF MODULE. SHOULD BE SCREW DRIVER ADJUSTABLE THROUGH SIDE PANEL, SO DRILL HOLES APROPRIATELY.

EXAMPLE: (R1T-R6T)



BLANKING & BURST AMP



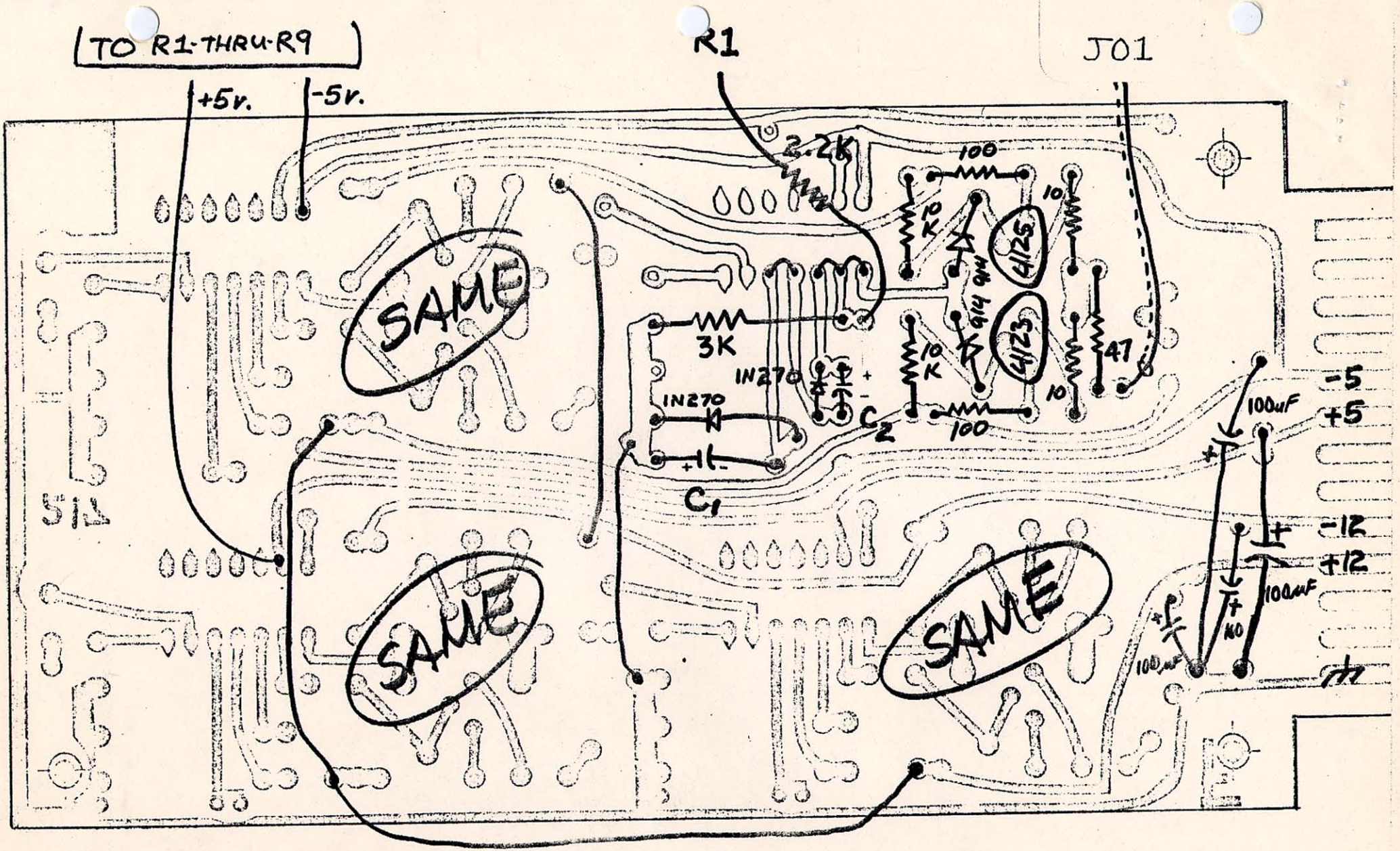
6		525-1060	10 OHM, 1/4 W RES	.06	A	IP
12		525-1165	75 OHM, 1/4 W RES	.06	A	IP
9		525-1202	100 OHM, 1/4 W RES	.06	A	IP
3		525-1248	220 OHM, 1/4 W RES	.06	A	IP
3		525-1270	300 OHM, 1/4 W RES	.06	A	IP
2		525-1467	5.1KOHM, 1/4 W RES	.06	A	IP
6		525-1522	10KKOHM, 1/4 W RES	.06	A	IP
6		525-1526	11KKOHM, 1/4 W RES	.06	A	IP
6		525-1717	100KOHM, 1/4 W RES	.06	A	IP
3		525-1302	470 OHM, 1/4 W RES	.06	A	IP
8		525-1421	2.7KOHM, 1/4 W RES	.06	A	IP
3		525-1433	3.3KOHM, 1/4 W RES	.06	A	IP
3		525-1650	47KKOHM, 1/4 W RES	.06	A	IP
3			150 OHM 1/4 W RES	.06	A	IP
6	10F655		50K TRIM,LOK,PNL-MT	3.00	N	IP
3	9F073	U1	100 OHM POT 1/4SFT	1.45	N	IP
3			.002 pF			
~~1~~	~~1F[illegible]~~	~~C[illegible]~~	~~15 PF,DIP-MICA CAP~~	~~.20~~	~~A~~	~~IP~~
6	14F592	DM15-417J	470 PF,DIP-MICA CAP	.32	N	IP
6	67F307	TOC-103Z	.01 MF,DSC-CER CAP.	.05	N	IP
6	71L-1251	2-6-101	10 MF,25VDC,ELEC.	.44	A	IP
6	71L-1260	7-6-1000	100 MF,25VDC,ELEC.	.24	A	IP
3			.003 pF			
8	353-0914	1N9146	SIL-DIODE	.19	A	IP
		1N270	GERM-DIODE	.22	S	IP
3		2N4123	NPN TRANS	.22	S	IP
3		2N4125	PNP TRANS	.27	S	IP
6		CA3030	DIP OP-AMP,RCA	1.32	S	IP
3	35F1956	SMD 39	39 MICROHENRY CHOKE	2.91	N	IP
3	35F1969	SMD470	470 MICROHENRY CHOK	2.91	N	IP
3			VSS P-C BOARDS			IP
16	39F1337	13-236	BNC,FM-CHS.MT.	.72	N	IP
1			CHASSIS,INPUT-FACE	8.25	DG	IPM
3	RB-57-1-DC-M-L-9		KNOB,BLACK/INLAY299	1.00	R	IP

REFERENCE MODULE:

The Reference module produces a constant voltage proportional to front panel knob position. It uses 2¼ #217 printed circuit boards; save other 3/4 of board for making 3-D Joystick later...

Joystick and slide pot inputs could be created in analogous manner. The value of input resistor, R1 through R9, is not critical; for instance if 5K ohm pots in joysticks are available, use them.

Capacitors C_1, C_2, are used to filter out noise. 100uF is the minimum and does not affect the feel much. Dan chose 250uF and Phil chose 1000uF; 1000uF is very 'slushy'.



CAPACITORS C_1, C_2, MAY BE ANY VALUE BETWEEN 100uf — 1000uf.

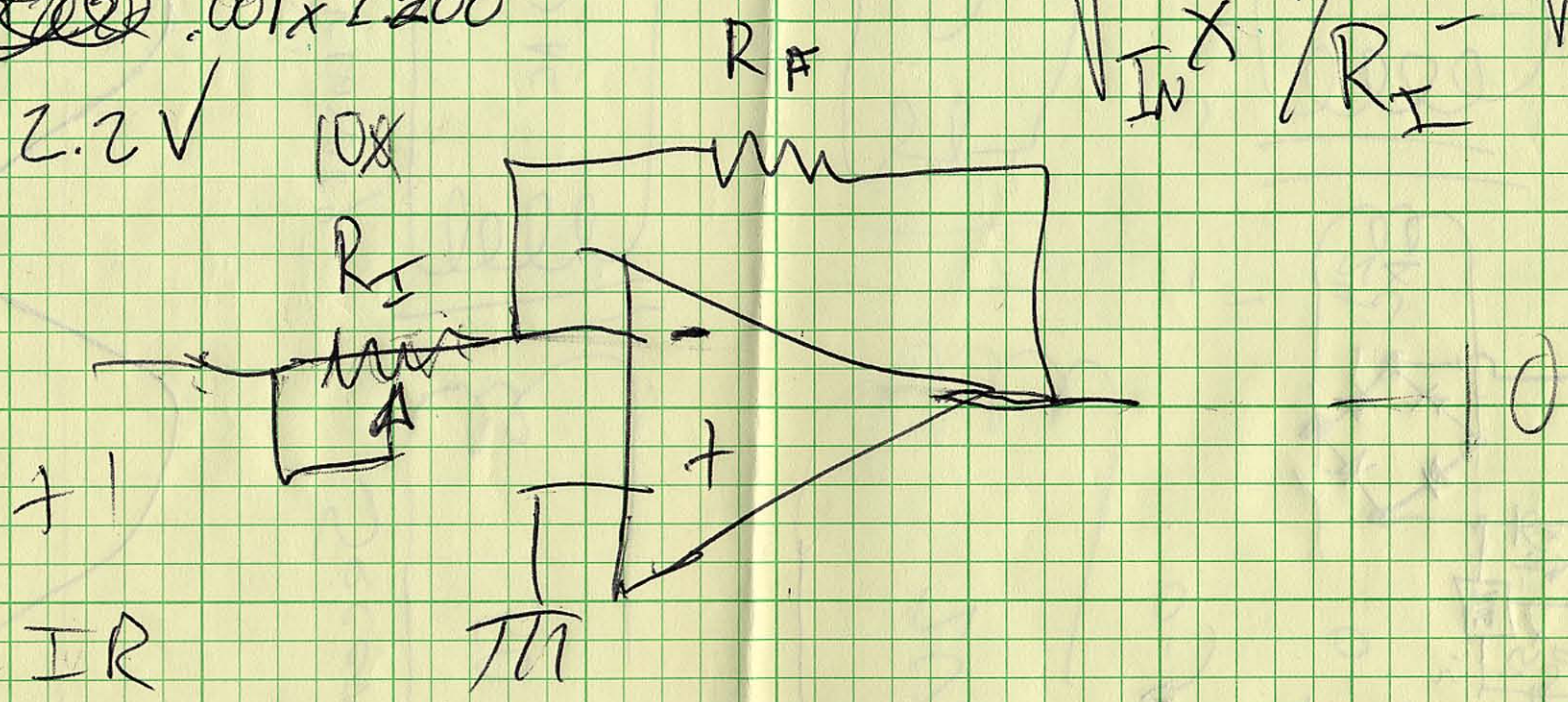
REFERENCE

$I = .0009615 \approx .001$ or $\frac{1mA}{100}$

$E = .001 \times 2200$

$E = 2.2V$ ✓

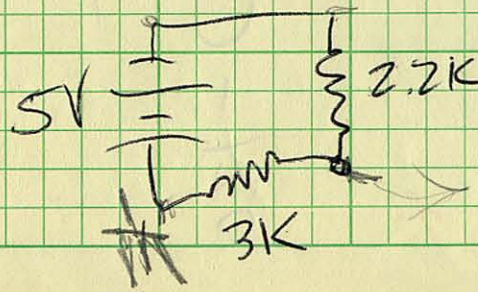
$V_{IN} \times R_F / R_I = V_{OUT}$



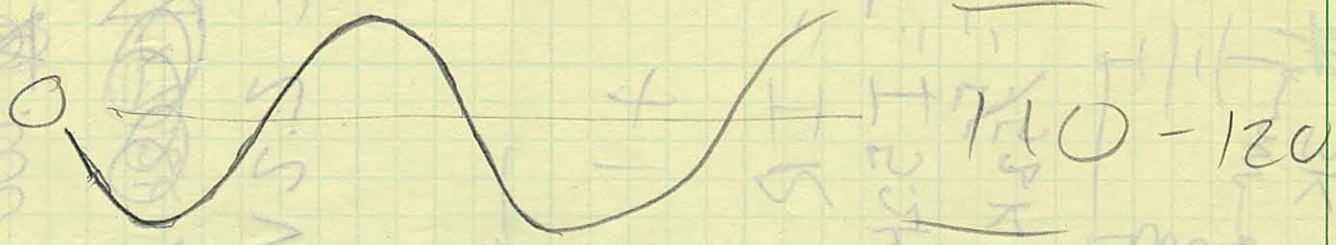
$E = IR$

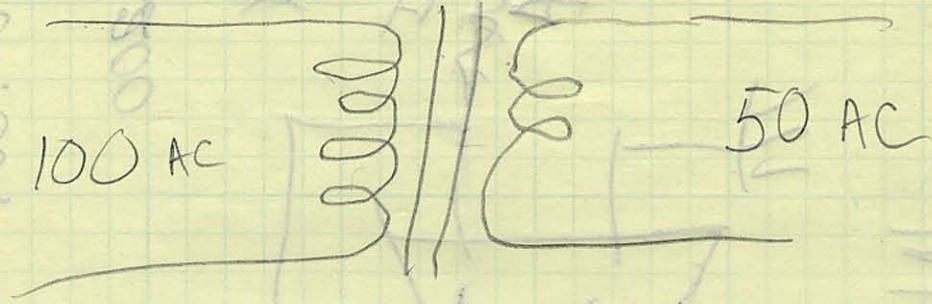
$5 = I\,5.2K$

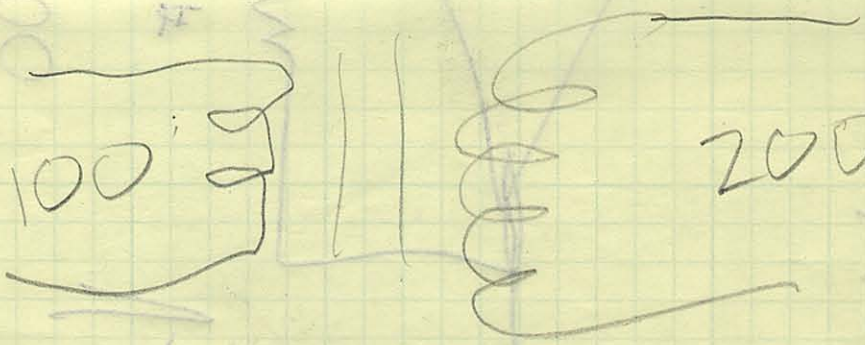
$I = 5/5.2K$

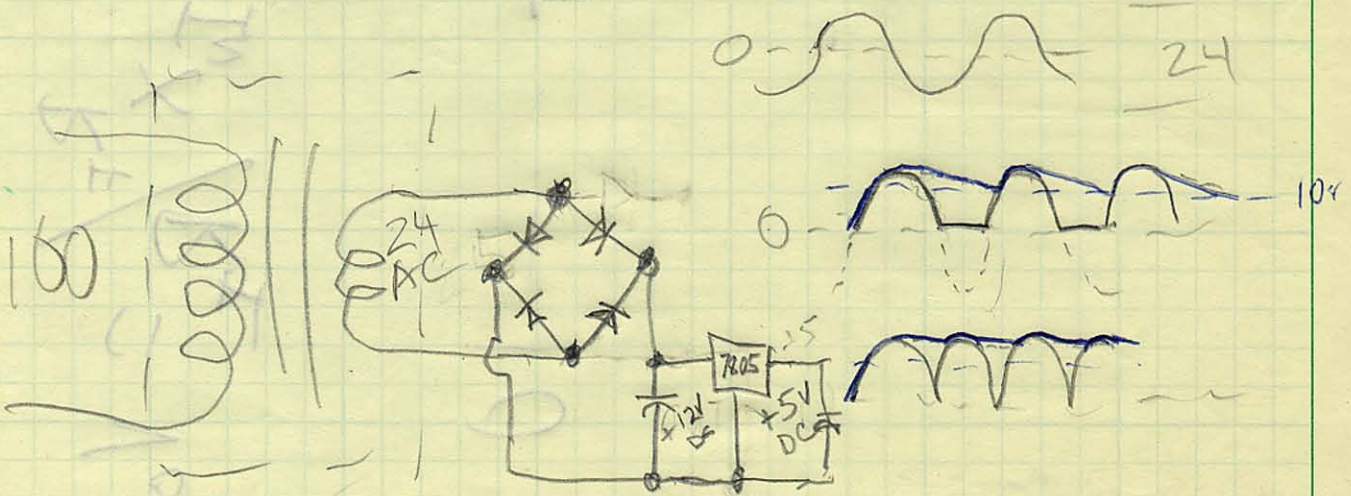


INVERTING OP AMP

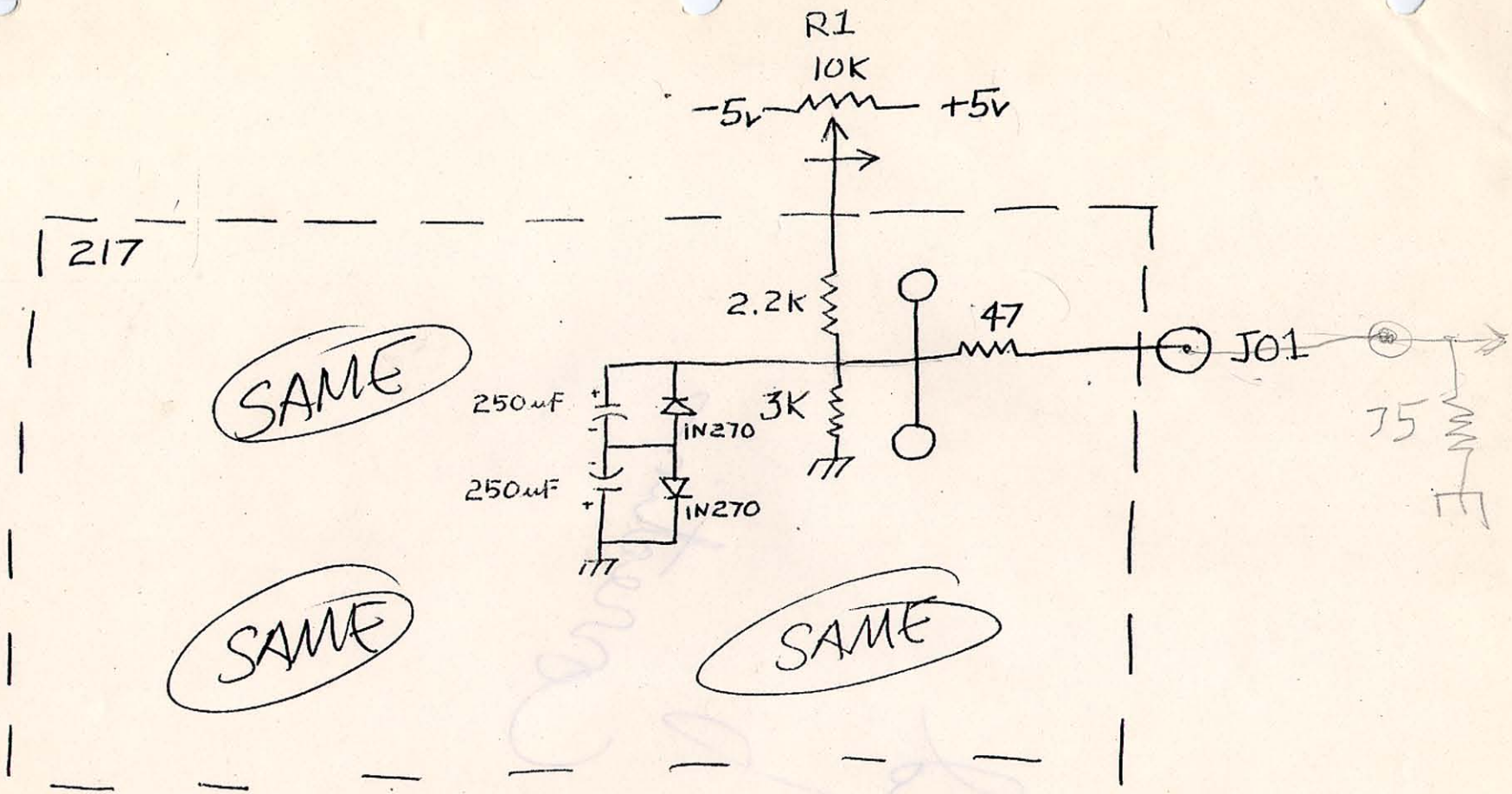


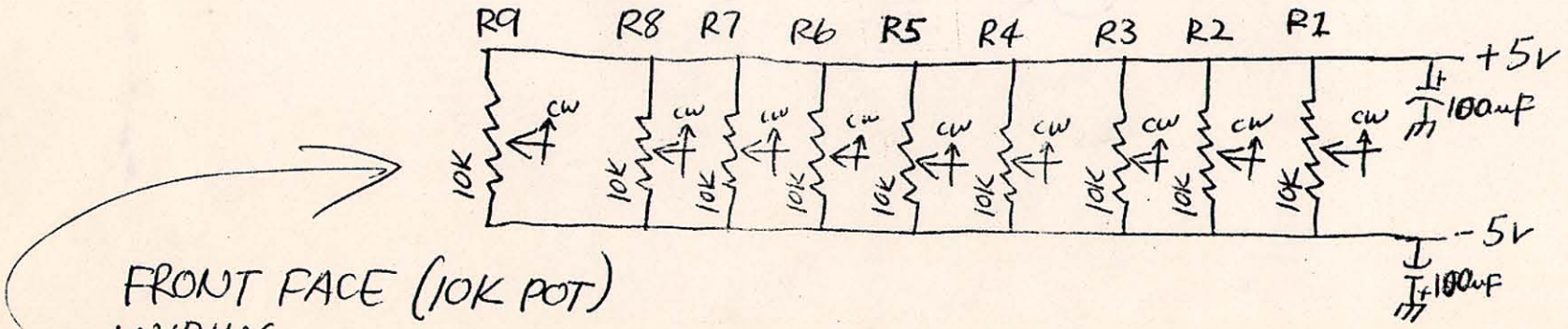






42-381 50 SHEETS 5 SQUARE
42-382 100 SHEETS 5 SQUARE
42-389 200 SHEETS 5 SQUARE
Made in U.S.A.
NATIONAL


R1
10K
-5v
+5v
217
2.2K
47
J01
SAME
250uf
3K
IN270
250uf
IN270
J5
SAME
SAME


R9
R8
R7
R6
R5
R4
R3
R2
R1
+5v
100uf
-5v
100uf
10K
CW
FRONT FACE (10K POT)
WIRING

REFERENCE

		525-1000	10 OHM, 1/4 W RES	.06	A	REF
4		525-1153	47 OHM, 1/4 W RES	.06	A	REF
10		525-1202	100 OHM, 1/4 W RES	.06	A	REF
9		525-1405	2.2KOHM, 1/4 W RES	.06	A	REF
9		525-3427	3K OHM, 1/4 W RES	.06	A	REF
10		525-1522	10KKOHM, 1/4 W RES	.06	A	REF
9	10F453	105UA	10K OHM POT 1/4SFAE	2.21	N	REF
4	710-1254	7-6-1000	100 MF,25VDC,ELEC.	.24	A	REF
10	583-0914	1N9148	SIL-DIODE	.19	A	REF
9		2N4123	NPN TRANS	.22	S	REF
9		2N4125	PNP TRANS	.27	S	REF
3			217 P-C BOARD			REF
2	59r1537	13-236	BNC,FM-CHS.MT.	.72	N	REF
1			CHASSISREFRENCE-FAC	8.25	DG	REF
9	83-67-1-0C-M-0-9		ANOB,BLACK/INLAY299	1.00	R	REF

18		1N270	GERM-DIODE	.22	S	REF
18		710-1260	100uF, 25v, ELECT.		A	REF
-or-						
18		710-1218	250uF, 12v, ELECT.		A	REF
-or-						
18		623-0701	470uF, 16v, ELECT.		A	REF
-or-						
18		623-0703	1000uF ELECT.		A	REF

AMPLITUDE CLASSIFIER

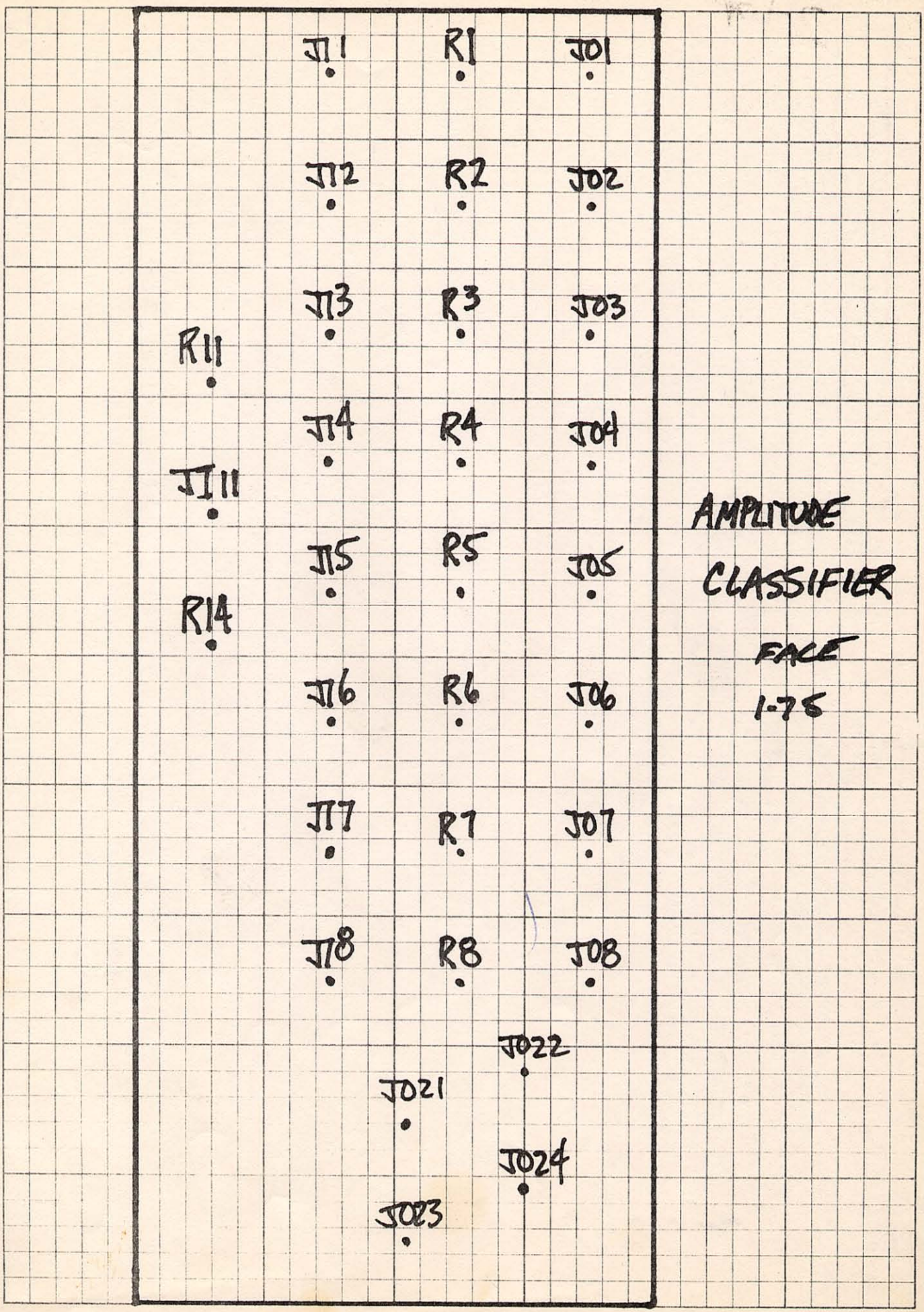
The amplitude classifier takes an input signal at JI 1land separates it into 8 contigeous regions varying from black to white. The value put out by each region is controled by R 1 through R 8 and by signals inputed to JI 1 through JI8. The output signals are available for each region seperately, JO 1 through JO 8. The sum of these signals is available at JO 21 through JO 24. The effect of JI 1, JI2, R 2, etc., is additive in each region. R11 controls the gain of the signal inputed at JI 11, and R 14 generates a bias (constant gray level proportional to knob position) which is added to the input signal. In general, R 11 and R 14 are used to match the incoming signal to the lightest, or 'top and 'darkest' or 'bottom' of the 8 regions.

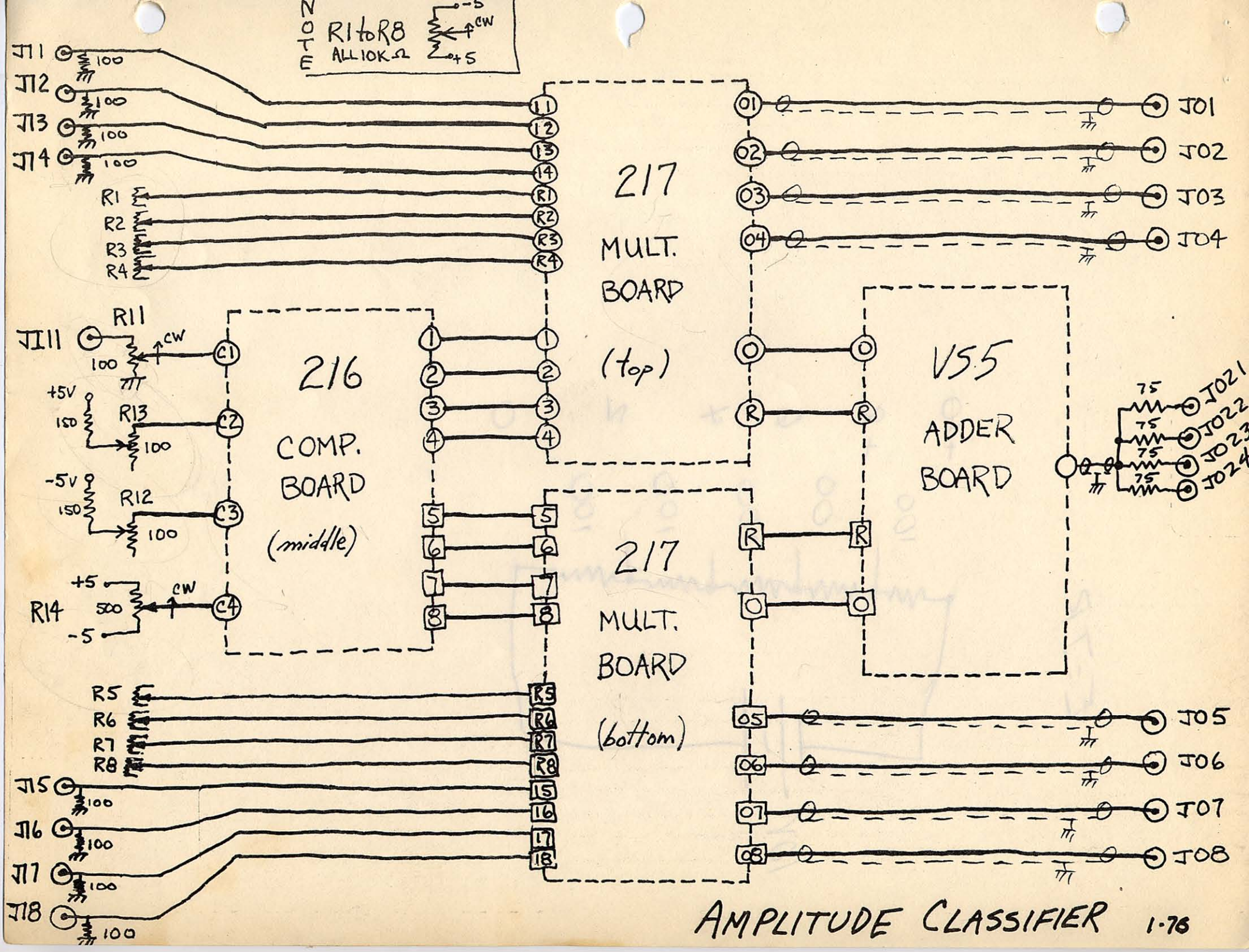
TEST INFORMATION

These tests are best performed with a 1 volt, peak to peak triangle wave inputed to JI 11, and a calibrated dual trace oscilloscope connected to the input and output of the amplitude classifier. R 13 is adjusted so that a +.4 volt signal activates channel 8 (bottom). R 12 is adjusted so that a +.4 volt signal activates channel 1 (top). R11 should be full clockwise and R 14 should be in the exact center of rotation. R 12 and R13 interact greatly, so, repeat adjustments until both conditions can be met simultaneously. R 16 should be adjusted so that with R 1 through R 8 in their centers the individual output are near 0 volts when not activated. R 15 should be adjusted so the summed output appearing at JO 21 through JO 24 is 0 volts for the non-activated channels.

BASIC CIRCUIT DESCRIPTION AND TEST INFORMATION--AMPLITUDE CLASSIFIER

A signal at JI11 is sent to all comparators. The resistor string of 100 ohm resistors combined with voltage sources at R 13 and R 12 bias so that IC 1 switches on at a higher (+) voltage than IC 2, IC 2 switches on at a higher voltage than IC 3, etc.. For example; with 0 volts at JI 11, IC 5,6,7 might be on and IC 1,2,3,4 would be off. The transistors connected to the outputs of the comparators decode the comparator string output such that only the highest comparator on is outputed. In this example, channel 5 would be on, all others off. This signal is sent to the multiplier associated with channel 5, turning it on. All other multipliers would be off.


JI1
R1
JO1
JI2
R2
JO2
JI3
R3
JO3
R11
JI4
R4
JO4
JI11
JI5
R5
JO5
R14
JI6
R6
JO6
JI7
R7
JO7
JI8
R8
JO8
JO22
JO21
JO24
JO23
AMPLITUDE
CLASSIFIER
FACE
1-76



AMPLITUDE CLASSIFIER 1-76